

BICHAT-BEAUJON YOUNG RESEARCHERS ASSOCIATION PRESENTS

Bichat-Beaujon Young Researchers DAY

OCTOBER 6TH 2026



PromoCell®

BECKMAN
COULTER
Life Sciences

BioLegend
From Revvity

proteintech®

fisher scientific
by Thermo Fisher Scientific

Novogene

ThermoFisher
SCIENTIFIC

MCE®
MedChemExpress

MANY THANKS TO OUR SPONSORS

JCB
JOURNÉE DES JEUNES CHERCHEURS
DE BICHAT-BEAUJON

CRI
CENTRE DE RECHERCHE
SUR L'INFLAMMATION



cnrs

iame

BIOCONVS
Bioconvergence pour la Santé

Domaine de Recherche et d'Innovation Majeur

FILIERE SANTÉ
MALADIES RARES
TeteCou
MALFORMATIONS RARES TÊTE COU DENTS

Inserm
La science pour la santé
From science to health

UNIVERSITÉ
SORBONNE
PARIS NORD

Université
Paris Cité

Région
île de France

The Association

The Young Researchers Association of Grand Paris Nord is a non-profit organization established under the French Law of 1901, founded in 2023. Its aim is to promote connections among engineers, postdoctoral researchers, PhD candidates and master's students from the laboratories of the future Grand Paris Nord Hospital, through the organization of several events.

JCBB
Since
2023

The Young Researchers Day of Bichat-Beaujon (JCBB) is our main event. It consists of a one-day seminar-conference during which young researchers present their work through oral presentations, short “pitch your project”-style flash talks in 180s, and poster sessions.

200+
Attendees in 2025

Our audience is primarily composed of students, contract researchers, research engineers and technical staff, as well as physicians from the invited laboratories. Initially organized for two years at the CRI (Centre de Recherche sur l'Inflammation) within the Faculty of Medicine, the event has expanded over the past three years to include all laboratories from the Bichat, Beaujon and Montrouge sites.

Contact

E-mail:

youngresearchersparisnord@gmail.com

LinkedIn :

Association des Jeunes Chercheurs de l'Hopital Grand Paris Nord

Co-organizing partners of the project :

Research units LVTS (INSERM U1148), CRI (INSERM U1149), IAME (INSERM U1137), U1152 & BRIO (URP2496)

Organization Team



Nicolas GODRON

IAME

PhD student - 3rd year
President / Pole Logistics

Elisa PERELLI

LVTS

PhD student - 3rd year
Tresorer / Pole sponsors



Emilie VONNA

LVTS

PhD student - 3rd year
Secretary / Pole communication

Louis BERTHET

CRI

PhD student - 4th year
Pole Logistics / New events



Elsa MEISSONNIER

LVTS

Research engineer
Pole communication

Margot AZOULAI

CRI

PhD student - 2nd year
Pole sponsors



Emma SULMONI

LVTS

PhD student - 2nd year
Pole sponsors

Organization Team



Jorge MARTINEZ-GARCIA

IAME

PhD student - 2nd year

Pole Logistics

Akanksha AGGARWAL

IAME

Research engineer

D-day Logistics



Mattéo LEGUEN

IAME

Research engineer

D-day Logistics

Agathe OK

CRI

PhD student - 1st year

D-day Logistics



Marguerite DUCAMP

CRI

PhD student - 4th year

D-day Logistic



Lunch



Figure 1



Faculty of medicine
Floor -1



Figure 1. The effect of the number of trials on the number of correct responses. The number of correct responses was significantly higher than the number of incorrect responses for all groups. The number of correct responses was significantly higher than the number of incorrect responses for all groups. The number of correct responses was significantly higher than the number of incorrect responses for all groups.



Sponsors



Main sponsor



Note that the floorplan might be subjected to changes.

Program

	08h00	Welcoming Coffee
	08h45	Opening Speech
	09h00	Plenary Session : Sustainability in Research Marianne De Paepe
	09h40	Oral Presentation - Session 1
	10h30	Poster - Session 1 (odd-numbered posters)
	11h15	Coffee & Sponsors break
	11h30	Oral Presentation - Session 2
	12h30	Lunch Break
	13h30	Oral Presentation - Session 3
	15h00	Plenary Session : From Scientific Discovery to Entrepreneurship Iman Hajar, Didier Letourneur, Pierre Launay
	16h00	Poster - Session 2 (even-numbered posters)
	16h45	Coffee & Sponsors break
	17h00	Pitch your project - 180s to present your project
	18h00	Jury Deliberation, Closing Speech & Award Announcement
	18h15	Closing Cocktail

Oral presentations

Session 1 / 9h40-10h30

Klaus HÄMÄLÄINEN (CRI)

The transcription factor Pax6 regulates microglial activation and restrains neuroinflammation

Paprapach WONGDONTREE (IAME)

AcrAB-TolC Efflux Pump Contributes to Drug- and Organ-Specific Antibiotic Tolerance in Uropathogenic Escherichia coli

Jessica TORMO (Santé Orale)

Effects of a Low-Dose Endocrine-Disrupting Chemical Mixture on the Lingual Mucosa: Epithelial and Neuronal Alterations in a Mouse Model

Session 2 / 11h30-12h30

Samuel CHAUVIN (CRI)

Renal senescence is associated with lesion development in monoclonal immunoglobulin deposition disease

Zhuo HUANG (LVTS)

CD31 signaling in chronic inflammatory diseases and fibrosis

Kérima IDRIS (Santé Orale)

Schwann cells regulate transcriptional and translational profiles of stemness markers in stem cells from human exfoliated deciduous teeth

Wiam GABEL (CRI)

Crosstalk between Primary Cilia and mitochondria in renal pathophysiology

Oral presentations

Session 3 / 13h30-15h00

Aurélie PHAN (CRI)

Pharmacogenetic study of the weight response to semaglutide 2.4 mg in severe obesity: from GLP1R rs6923761 to an 8-gene NGS panel

Laura GIORDANO (CRI)

Fecal let-7b and miR-21 directly impact intestinal microbiota in IBD

Giang Ngoc Truong NGUYEN (IAME)

Modelling the Impact of Antiviral Therapy, Oral and Long-Acting Injectable Pre-Exposure Prophylaxis on HIV Incidence and Mortality among Sexually Active Men who have Sex with Men in metropolitan France (1972–2030)

Morgane DE SOUSA (LVTS)

Optogenetic nanotool to investigate transcription factor activity

Maxime AMOYEL (CRI)

Role of hormonal and microbiome modulation on gastric emptying and metabolic response after endoscopic sleeve gastropasty (HOVIGAMI-Endosleeve): a prospective comparative study

Pauline CHAUMET (IAME)

Nationwide digital surveillance reveals resurgence of pertussis through real-time decision support systems

Posters

odd n° = Session 1

even n° = Session 2

Ghalia BEN GHEDIFA (CRI)

1

A Multimetric Framework for Standardized Assessment of Mass Spectrometry Imaging Protocols: Hepatocellular Carcinoma as a Proof of Concept

Noémie RYBAK (IAME)

2

Deciphering the catalytic mechanisms and substrate specificity of *Klebsiella pneumoniae* carbapenemase using site-directed mutagenesis and deep mutational scanning

3

Mattéo LEGUEN (IAME)

The resistome of *Enterococcus*

4

Louis MARSTRAND-DAUCE (CRI)

A high-fat diet is insufficient to drive pancreatic intraepithelial neoplasia from acinar-to-ductal metaplasia

Angélique BIBIMBOU (LVTS)

5

Etude du syndrome de Myhre : Caractérisation du phénotype cardiaque et pulmonaire du modèle murin *Smad4* Ile500/+

Lucas BODELLE (IAME)

6

Multi-response extension and computational optimisation of a framework coupling non-linear mixed-effects modeling with whole-body PBPK models

7

Rime CHEBBO (IAME)

Deciphering the role of the recently identified *eefRABCD* efflux pump in *E.coli*

8

Nathan HENNION (CRI)

Development of a pulmonary fibrosis model in immune-humanized BRGSF mice

Posters

odd n° = Session 1

even n° = Session 2

- 9** Linkang MENG (CRI)
Epigenetic control of HDAC3: From Pancreatic Regeneration to Tumor Initiation
- 10** Adeline DOS SANTOS (LVTS)
Magneto-Plasmonic Polysaccharide Particle-Assisted Thrombolysis
- 11** Mélanie GARMON (CRI)
Le tirzépatide remodèle l'identité transcriptomique des cellules caliciformes dans le jéjunum d'un modèle de rat de diabète de type 2
- 12** Agathe OK (CRI)
Loss of one of the isoforms of the mesenchymal transcription factor PRRX1 in pulmonary fibrosis associated with systemic sclerosis: a new pathophysiological mechanism?
- 13** Martin ZOUZOU (CRI)
Role of Lcn2 in mast cells activation: Pathological implications in chronic kidney disease
- 14** Muayad AL HAFEZ (LVTS)
Development of a preclinical model of carcinoid heart disease (CaHD)
- 15** Laura MULAS (IAME)
Characterizing viral load dynamics in community SARS-CoV-2, Influenza, and RSV infections : the role of age, sex, and vaccination

Posters

odd n° = Session 1

even n° = Session 2

Célia SCRIBE (CRI)

16

Lung adipose metaplasia contributes to a fibrogenic and inflammatory microenvironment in Interstitial Lung Diseases (ILD)

Aglaé PERRIN (IAME)

17

Genome wide association study of model-derived time to viral clearance in COVID-19: comparison of methods in a simulation study and application to the discovery trials

Chengli YU (IAME)

18

Usage and regulatory acceptance of PBPK models in paediatric drug applications: a review of EMA and FDA public assessment reports

Pitch your project

17h00-18h00

Berliana NOVIANITA (LVTS)

Activable Antioxidant Microbubbles for the Treatment of Marginal Organs via Ex Vivo Perfusion

Iris BOBINNEC (CRI)

Study of the immunomodulatory role of IL-1RA in Pulmonary Fibrosis

Jee-Seon YANG (IAME)

Invasive pneumococcal disease dynamics among sickle cell disease children and potential benefit of next-generation pneumococcal conjugate vaccines: a 15-year national cohort study

Arantxa VAQUERO (LVTS)

Development of a small perfusable artery *in vitro*

Lucy BOURLET (CRI)

Understanding the mechanisms of resistance to targeted internal radiotherapy in neuroendocrine tumors, and improving its efficacy

Jorge MARTINEZ-GARCIA (IAME)

Less bugs, more signal: reporter species as microbiome biomarkers

Emma SULMONI (LVTS)

A micropatterned hydrogel platform for human myocardium tissue modeling

Oral Presentations

ABSTRACTS

The transcription factor Pax6 regulates microglial activation and restrains neuroinflammation

Klaus HÄMÄLÄINEN (1), Marie D'ALLEMAGNE (1), Etienne BECHT (1), Cosmin SAVEANU (2), Daniel ABERDAM (3), Irini EVNOUCHIDOU (1), Loredana SAVEANU (1)

(1) Centre de Recherche sur l'Inflammation (CRI), INSERM U1149, CNRS ERL8252, Université Paris Cité, France. (2) Institut Pasteur, Université Paris Cité, France. (3) Centre de Recherche des Cordeliers, INSERM U1138, Université Paris Cité, France.

Microglia are Central Nervous System (CNS)-resident macrophage-like cells that modulate neurodevelopment and maintain neural homeostasis by sensing changes in the CNS. Therefore, their inflammatory responses must be tightly controlled. Uncontrolled activation of these cells can disrupt myelin, compromise neural function, and contribute to neurodegenerative diseases. This underscores the importance of identifying mechanisms that modulate microglial inflammation. Despite the significance of PAX6 as a key transcription factor in neurogenesis, its role in microglia remains uninvestigated.

We demonstrated that PAX6 expression is upregulated in activated microglia, supporting the hypothesis that PAX6 limits microglial inflammatory responses. To investigate its role in CNS inflammation, we generated mice with microglia-specific Pax6 deletion, as well as a human microglia cell line depleted of PAX6. Using these models, we demonstrated that loss of PAX6 shifts microglia toward a pro-inflammatory phenotype, characterized by enhanced inflammatory and type I interferon signatures, and that Pax6 deficiency exacerbates disease severity in experimental autoimmune encephalomyelitis (EAE), a mouse model of multiple sclerosis (MS).

To gain mechanistic insight, we performed antibody-targeted chromatin profiling (CUT&RUN) in human and mouse microglia. This analysis revealed PAX6 binding to genes involved in the nonsense-mediated mRNA decay (NMD) pathway, which is an RNA surveillance mechanism that limits the abundance of specific transcripts by promoting their degradation. Our results indicate that PAX6 controls the inflammatory response by promoting the degradation of inflammatory transcripts through NMD. By identifying PAX6 as a regulator of NMD-dependent inflammatory gene expression, this work reveals a previously unrecognized mechanism that controls microglial inflammation and contributes to our broader understanding of immune gene regulation.

AcrAB-TolC Efflux Pump Contributes to Drug- and Organ-Specific Antibiotic Tolerance in Uropathogenic *E. coli*

Paprapach WONGDONTREE (1), Sara DION (1), Imane EL-MEOUCHE (1)

(1) Université Paris Cité, Université Sorbonne Paris Nord, Inserm, IAME, 75018 Paris, France.

Urinary tract infections (UTIs) are a major global health burden, affecting both men and women worldwide. A major clinical challenge of UTIs is the high rate of relapse and recurrence following antibiotic treatment. Increasing evidence suggests that these treatment failures may be linked to bacterial antibiotic tolerance, a transient physiological state that allows bacteria to survive antibiotic exposure without necessarily acquiring genetic resistance. *Escherichia coli*, particularly uropathogenic *E. coli* (UPEC), is the leading cause of the infection. Although the AcrAB-TolC efflux pump is a well-established determinant of antibiotic resistance in *E. coli*, its contribution in antibiotic tolerance during UTIs remains poorly understood. Here, we investigated the contribution of AcrAB tolerance in diverse clinical UPEC isolates using in vitro, ex vivo, and in vivo mouse infection models. Inactivation of *acrAB* did not affect UPEC infection capacity at day 2 post-infection in mouse model, suggesting that AcrAB is not required for early urinary tract colonization. However, ex vivo antibiotic efficacy revealed strongly organ- and drug-dependent: ciprofloxacin was more effective in the kidney, whereas delafloxacin was more effective in the prostate. Notably, *acrAB* deletion further enhanced bacterial killing by delafloxacin in the prostate and by ciprofloxacin in the kidney. In contrast, although *acrAB* deletion increased ampicillin susceptibility in vitro, this effect was absent ex vivo of kidney and prostate, suggesting that tissue-associated factors modulate AcrAB-dependent tolerance. Together, these findings identify AcrAB as a selective, rather than universal, determinant of antibiotic tolerance and highlight drug-niche interactions as an important consideration for UTI treatment.

Effects of a Low-Dose Endocrine-Disrupting Chemical Mixture on the Lingual Mucosa: Epithelial and Neuronal Alterations in a Mouse Model

Jessica TORMO (1), Ines BELKADI[†], Sophia LOIODICE (2), Siwar RICH (3), Coralie TORRENS (1), Asmaa FODA (1), Ali NASSIF (1)(4), Catherine CHAUSSAIN (1)(5), Jean-Marc RICORT (1)(6), Yves BOUCHER (1)(4), Sylvie BABAJKO (1), Ihsène TAIHI (1)(7)

(1) Université Paris Cité, UFR Odontologie, INSERM 1333 UMR Santé Orale, Montrouge, France. (2) Biological Resource Platform (PRB), Pitié-Salpêtrière University Hospital, Paris, France. (3) Research Laboratory on Biocompatible Substances, Faculty of Dental Medicine of Monastir, Higher Institute of Biotechnology of Monastir, Tunisia. (4) Department of Dentistry, Pitié-Salpêtrière Hospital, Assistance Publique-Hôpitaux de Paris, France. (5) Department of Dentistry, Bretonneau Hospital, Assistance Publique-Hôpitaux de Paris, France. (6) Université Paris Saclay, France. (7) Department of Oral Surgery, Rothschild Hospital, Assistance Publique-Hôpitaux de Paris, France.

[†]Contributed as a co-first author.

The tongue, a key interface between sensory and digestive functions, remains relatively understudied in environmental pathophysiology, despite its continuous exposure to environmental and dietary factors. This project investigates the effects of chronic exposure to a low-dose mixture of 15 endocrine-disrupting chemicals (EDCs) on the lingual mucosa in mice, using complementary *in vivo* and *in vitro* approaches. Male and female mice were exposed *in utero* through to 10 months of age to a mixture of 15 EDCs, allowing the effects of prolonged, low-dose exposure to be investigated across development and adulthood. Tongue tissues were subsequently analysed by immunofluorescence to assess nerve fiber density and epithelial proliferation, enabling both morphological and neuronal alterations to be characterised.

Our findings reveal significant alterations in the lingual epithelium of male mice following 10 months of low-dose EDC exposure, consistent with changes previously reported in high-dose exposure models. Notably, these affected regions exhibited a pronounced increase in epithelial mitotic activity, indicative of enhanced cellular proliferation and suggesting disruption of normal epithelial homeostasis. We also observed a trend towards reduced intraepithelial nerve fiber density in the anterior tongue, where innervation is normally highest. This reduction may indicate alterations in the sensory component of the lingual mucosa and suggests that neuronal structures may also be sensitive to chronic low-dose environmental exposure.

Together, these findings support the hypothesis that chronic exposure to low-dose EDC mixtures can disrupt oral mucosal homeostasis, with effects differing by sex. The observed epithelial and neuronal alterations highlight the potential sensitivity of the tongue to environmental endocrine disruption and provide evidence that prolonged low-dose exposure may produce measurable tissue-level effects. By identifying potential early biomarkers of exposure and characterising epithelial and neuronal alterations, this study advances our understanding of how environmental pollutants may affect the oral mucosa and provides a foundation for further investigation into their consequences for oral health.

Renal senescence is associated with lesion development in monoclonal immunoglobulin deposition disease

Samuel CHAUVIN (1), Thomas LOEFFLER (5), Ariane COUTROT (2), Léa CHAPART (2), Sébastien BENDER (3), Christophe SIRAC (3), Camille COHEN (4)

(1) INSERM U1149, Centre de Recherche sur l'Inflammation, Université Paris Cité, Paris, France. (2) INSERM U1149, Centre de Recherche sur l'Inflammation, Paris, France. (3) CRIBL Unit, UMR CNRS 7276 INSERM 1262, University of Limoges, Limoges, France. (4) INSERM U1149, Centre de Recherche sur l'Inflammation, Université Paris Cité, Nephrology Department, Bichat Hospital, APHP, Paris, France. (5) INSERM U1149, Centre de Recherche sur l'Inflammation, Department of Nephrology, Centre Hospitalier Régional Universitaire de Nancy, France.

Background

Monoclonal Immunoglobulin Deposition Disease (MIDD) is characterized by the deposition of pathogenic monoclonal immunoglobulins (MIg) in the kidney, leading to tubulointerstitial fibrosis and glomerulosclerosis. The mechanisms linking deposits to kidney lesions remain poorly understood. Cellular senescence—a state of stable cell-cycle arrest associated with a senescence-associated secretory phenotype (SASP)—has been implicated in glomerulosclerosis of other causes. Here, we identify an upregulation of cellular senescence in MIDD, associated with the development of glomerulosclerosis and the recruitment of Trem2⁺ macrophages, which may drive lesion progression.

Methods

We combined the analysis of human MIDD kidney biopsies with transgenic mice recapitulating MIDD lesions (10.1182/blood.2020005980). Tissue characterization was performed using immunofluorescence, complemented by high-resolution spatial transcriptomics.

Results

In both mouse and human kidney tissues, we observed a marked accumulation of senescent cells, evidenced by the presence of p21-positive nuclei in both glomerular and tubular epithelial cells. Spatial transcriptomic analysis revealed that these senescent glomeruli exhibited a pronounced upregulation of genes associated with the matrisome, consistent with an active fibrotic process and sclerosis. Further analysis indicated a selective enrichment of myofibroblasts, key effectors of fibrosis, and Trem2⁺ macrophages, a subset associated with tissue remodelling and chronic injury, around senescent glomeruli, suggesting potential paracrine interactions.

Conclusion

These findings support a role for cellular senescence in MIDD-associated renal injury. As senescence induces SASP, we hypothesize that SASP factors recruit macrophages, which in turn promote fibrosis and glomerulosclerosis. Ongoing work aims to determine whether targeting senescent cells with senolytic can mitigate lesion development, opening new therapeutic perspectives in MIDD.

CD31 signaling in chronic inflammatory diseases and fibrosis

Zhuo HUANG (1), Marc CLEMENT (1), Giuseppina CALIGIURI (1), Charles CHEVILLARD (1)

(1) Team 1, LVTS UMRS 1148 INSERM, France.

Chronic inflammatory diseases (glomerulonephritis, rheumatoid arthritis...) are major health burden world-wide, that finally lead to fibrosis and loss of organ function. In the progression of these diseases, chronic vascular inflammation plays an important role, which is characterised by endothelial cell activation, platelet recruitment and leukocyte infiltration. Therefore, studying the interaction between these cells are crucial to the understanding of chronic vascular inflammation. CD31 is a transmembrane immunoglobulin-like protein that is widely expressed on all these 3 types of cells, which has shown regulatory effect in cellular interaction. Hence we hypothesise that CD31 signaling can regulate chronic vascular inflammation, and eventually regulate inflammation and fibrosis.

In this study, 3 series of tissue-specific CD31 signaling KO mice were used (endothelial cell KO, myeloid cell KO, platelet KO), which was achieved using the Cre-loxP recombination system. An antibody-mediated glomerulonephritis model was done by injecting nephrotoxic serum i.v. for the study of inflammation and fibrosis.

With the KO of CD31 signaling in myeloid cells, we observed no difference of disease phenotype compared to control group. However, with the KO of CD31 signaling in endothelial cells, the mice showed worsened glomerular barrier function, worse kidney insufficiency, stronger systemic inflammation and more tissue remodelling. For the moment, platelet CD31 signaling KO model is not experimented on for the lack of mice.

These results indicate that CD31 signaling in endothelial cells protects kidney during antibody-mediated glomerulonephritis. Furthermore, this CD31 signaling KO model will allow the study of inflammation and fibrosis in many other diseases, driving the potential to new molecular mechanism and therapeutic targets.

Schwann cells regulate transcriptional and translational profiles of stemness markers in stem cells from human exfoliated deciduous teeth

Kérïma IDRÏSS (1)

(1) Laboratoire de Santé Orale UMR1333, Université Paris Cité, UFR d'Odontologie.

Schwann cells (SCs) are glial cells of the peripheral nervous system that are essential for axonal maintenance and regeneration. SCs are components of stem cells niche and seem to play an important role in angiogenesis and osteogenesis, implying a broader role in tissue regeneration. Stem cells from human exfoliated deciduous teeth (SHEDs) exhibit high proliferative and differentiation potential and display robust pro-angiogenic and pro-osteogenic properties. Our aim is to evaluate the impact of SCs on SHEDs' stemness and osteogenic potential. In order to assess SCs' impact on SHEDs under direct contact, we produced spheroids of SCs-only, SHEDs-only or SHEDs+SCs (1:1 proportion) and cultured them in basal or osteogenic conditions for 7 days. Histological and immunofluorescence analyses showed that SHEDs+SCs spheroids were bigger and less dense than SHEDs-only spheroids with SHEDs predominantly localized in the spheroid's core while SCs were at the periphery. Compared with SHEDs-only spheroids, cocultured SHEDs (SHEDs-co) exhibited reduced apoptosis (nuclear Caspase 3 expression) and lower proliferation (nuclear PCNA expression). After dissociating the spheroids, gene expression of each cell type was analyzed. In basal conditions, SHEDs-co had a panel of stemness markers (OCT4, SOX2, NANOG, and KLF4) upregulated at a gene level relative to SHEDs-only. At a protein level, no differences were detected in NANOG level, SOX2 was upregulated, while OCT4 was downregulated. Under osteogenic conditions, no differences were observed at gene level for KLF4, and the early osteogenic markers COL1A1 and RUNX2 were downregulated in SHEDs-co relative to SHEDs-only, while no differences were detected for ALPL and BGLAP. At a protein level, OSX was downregulated in SHEDs-co relative to SHEDs-only. A different scenario was observed when investigating the paracrine effect of SCs on SHEDs. Although stemness-related gene expression was not significantly altered, SCs' secretome-treated SHEDs displayed increased nuclear expression of OCT4, SOX2, and NANOG. After 7 additional days of osteogenic treatment, treated SHEDs exhibited decreased expression of the early osteogenic marker RUNX2. Similarly, after 21 days of induction, the expression of the late osteogenic marker BGALP was reduced in treated SHEDs and conversely, treated SHEDs showed enhanced mineralization. In parallel, we sought to determine the molecules through which SCs impact SHEDs' stemness markers. Gene expression of functional candidate markers (VEGFA, CNTF, PDGFA, and PTH) was assessed in SCs co-cultured with SHEDs (SCs-co). No differences were detected for VEGFA and PTH, while CNTF and PDGFA were downregulated in SCs-co relative to SCs-only. Our preliminary findings suggest that SCs regulate SHEDs' stemness through direct contact while the paracrine effects of SCs do not produce the same results. Further analyses are needed to clarify how SCs affect SHEDs' osteogenic potential.

Crosstalk between Primary Cilia and mitochondria in renal pathophysiology

Wiam GABEL (1)

(1) Université Paris Cité, BioSpc.

Les dysfonctionnements mitochondriaux jouent un rôle central dans le développement des lésions rénales au cours de l'insuffisance rénale chronique. Si l'influence de la fonction mitochondriale sur le cil primaire a récemment été démontrée, le rôle du cil primaire dans le maintien de l'homéostasie mitochondriale reste largement méconnu. Nous avons donc émis l'hypothèse que la perte du cil primaire altère la dynamique et la fonction mitochondriales, contribuant ainsi au développement des lésions de l'épithélium tubulaire rénal.

Afin de tester cette hypothèse, nous avons utilisé des cellules épithéliales rénales MDCK présentant un knock-down inducible de KIF3A ou IFT88, deux gènes essentiels à la ciliogenèse. L'analyse par microscopie confocale révèle que l'absence de cil primaire induit une fragmentation mitochondriale, caractérisée par des mitochondries plus petites et plus arrondies, associée à une diminution de leur activité. Ces altérations s'accompagnent d'une diminution de l'expression de SIRT3, déacétylase majeure impliquée dans l'homéostasie mitochondriale, d'une augmentation de l'ATF5, marqueur de la réponse au stress mitochondrial (UPRmt), ainsi que d'une inhibition de l'autophagie et d'une activation de la voie mTOR.

Afin d'identifier les mécanismes responsables de ce phénotype, nous avons étudié le rôle de LKB1, une kinase localisée au niveau du cil primaire. Son knock-down ne reproduit ni les altérations morphologiques mitochondriales ni les modifications des voies de signalisation observées après perte du cil primaire, suggérant que LKB1 n'est pas impliquée dans ce processus. En revanche, l'inhibition pharmacologique de la voie mTOR par la rapamycine restaure la morphologie mitochondriale. Des expériences complémentaires utilisant un inhibiteur pharmacologique de l'autophagie montrent que cet effet est dépendant de la voie mTOR, mais indépendant de l'autophagie, confirmant ainsi le rôle majeur de cette voie dans les altérations mitochondriales induites par la perte du cil primaire.

Enfin, les premières analyses réalisées in vivo dans des modèles murins invalidés pour KIF3A ou IFT20, deux gènes indispensables à la ciliogenèse, confirment ces observations en mettant en évidence une diminution de la taille des mitochondries dans les cellules épithéliales rénales. Dans leur ensemble, ces résultats mettent en évidence un rôle essentiel du cil primaire dans le maintien de l'homéostasie mitochondriale et identifient la voie mTOR comme un acteur clé des dysfonctionnements mitochondriaux induits par sa perte. Ils ouvrent ainsi de nouvelles perspectives pour comprendre les mécanismes impliqués dans le développement des lésions rénales et identifier de nouvelles cibles thérapeutiques.

Pharmacogenetic study of the weight response to semaglutide 2.4 mg in severe obesity: from GLP1R rs6923761 to an 8-gene NGS panel.

Aurélié PHAN (1), Claire CARETTE (1), Céline NARJOZ (2), Nicolas PALLET (3), Claire RIVES-LANGE (4), Nathalie RASSY (5), Sébastien CZERNICHOW (4)

(1) Nutrition Department, Assistance Publique-Hôpitaux de Paris, Hôpital Européen Georges Pompidou, PIMS team, Center for Research on Inflammation, Inserm UMRS 1149, Paris, France. (2) Biochemistry Department, Assistance Publique-Hôpitaux de Paris, Hôpital Européen Georges Pompidou, Paris, France. (3) Biochemistry Department, Assistance Publique-Hôpitaux de Paris, Hôpital Européen Georges Pompidou, Cordeliers Research Center, MEPPOT team, Inserm UMRS 1138, Paris, France (4) Nutrition Department, Assistance Publique-Hôpitaux de Paris, Hôpital Européen Georges Pompidou, Center for Research in Epidemiology and Statistics, METHODS Team, Inserm UMR 1153, Paris, France. (5) Nutrition Department, Assistance Publique-Hôpitaux de Paris, Hôpital Européen Georges Pompidou, Paris, France.

Introduction

Obesity is a chronic disease affecting up to 18% of the French population and 13% of adults worldwide. It is associated with several metabolic, mechanical and cardiovascular disorders and shortens life expectancy at its most severe stage ($\text{BMI} \geq 40 \text{ kg/m}^2$). GLP1 receptor agonists (GLP1-RA), liraglutide and semaglutide, have radically changed the pharmacological treatment of obesity, allowing 8 to 15% weight loss in large randomized trials (SCALE and STEP programs), while dual GLP1/GIP receptor agonists such as tirzepatide can achieve up to 20% weight loss after one year. However, whatever the molecule used, the weight response is highly variable, ranging from non-responders ($< 5\%$ weight loss) to very good responders exceeding 15%, both in clinical trials and real-world settings. Beyond clinical determinants such as type 2 diabetes, age and sex, pharmacogenetics may account for part of this heterogeneity. The GLP1R rs6923761 variant (c.502G>A), found in 29% of the Caucasian population, has been associated with a poorer glycaemic response to DPP-4 inhibitors and GLP1-RA, but with a greater weight response to liraglutide in carriers of the minor A allele.

Methods

We conducted a feasibility study in a cohort of individuals with severe obesity ($\text{BMI} \geq 40 \text{ kg/m}^2$) treated with semaglutide 2.4 mg/week and genotyped for rs6923761. Participants were enrolled through the first French Early Access Programme for semaglutide 2.4 mg, granted to people living with class 3 obesity ($\text{BMI} \geq 40 \text{ kg/m}^2$) and at least one obesity-related comorbidity, and were followed up in the Specialized Obesity Centre of the European Hospital Georges Pompidou. We collected patient data at treatment initiation and at the first follow-up visit after prescription. The main outcome was the percentage of weight loss according to genotype and multivariable models including sex and age.

Results

We collected data for 112 patients including 73 women, with a mean age of 50.2 years (± 15.1), a mean BMI of 47.4 kg/m^2 (± 7.4) and a mean follow-up duration of 7.3 months (± 3). We showed that AA homozygotes lost weight significantly faster than AG/GG individuals ($1.64 \pm 0.78 \text{ \%/month}$ vs $1.04 \pm 0.79 \text{ \%/month}$; $p = 0.03$). Both genotype and female sex were independently associated with the rate of weight loss in a multiple regression analysis ($p = 0.045$ and $p = 0.0004$ respectively). These findings provide the first evidence that GLP1R rs6923761 modulates the kinetics of the early weight response to semaglutide 2.4 mg, and support the feasibility of a larger pharmacogenetic investigation.

Future Direction

We have therefore extended the approach by next-generation sequencing of an eight-gene panel across the whole cohort: GLP1R, ARRB1, ARRB2, DPP4, GIPR, TCF7L2, CNR1, and SORCS1. Candidate genes were selected based on GLP1 receptor signaling biology and previous reports of associations with clinical response to GLP1-RA. Associations will be analyzed using data collected every 6 months over a two-year period, allowing assessment of both the magnitude and the long-term trajectory of the weight response.

Fecal let-7b and miR-21 directly impact intestinal microbiota in IBD

Laura GIORDANO (1)(2), Maryline ROY(1), Maeva DUQUESNOY (3), Quentin LAMY-BESNIER (3), Héloïse RYTTER (3), Benoit CHASSAING (2)(3), Djésia ARNONE (2)(4), Laurent PEYRIN-BIROULET (4), Servane ALIROL (1), Jean-Pierre HUGOT (1), Émilie VIENNOIS (1)(2)

(1) INSERM U1149, Centre de Recherche sur l'Inflammation, Université Paris Cité, Paris, France. (2) INFINY Institute, Nancy University Hospital, Vandœuvre-lès-Nancy, France. (3) Microbiome-Host Interactions, Institut Pasteur, Université Paris Cité, INSERM U1306, CNRS UMR6047, Paris, France. (4) Department of Gastroenterology and Inserm NGERE U1256, University Hospital of Nancy, University of Lorraine, Vandoeuvre-les-Nancy, France.

Background

The intestinal microbiota plays a central role in maintaining intestinal homeostasis and is increasingly recognized as an important contributor to the pathogenesis of inflammatory bowel disease (IBD). MicroRNAs (miRNAs) have recently emerged as potential modulators of the microbiota, with evidence suggesting that they can interact directly with bacterial communities and influence both microbial composition and function. In the context of IBD, we previously identified miR-21 and let-7b as two miRNAs capable of modulating host-microbiota interactions, with let-7b altering intestinal microbiota composition and miR-21 contributing to impairment of the intestinal barrier. We therefore hypothesized that miR-21 and let-7b are associated with specific bacterial species in patients with IBD and could directly modulate bacterial physiology and transcriptional activity.

Methods

To investigate associations between host miRNAs expression and microbial composition, we performed multivariable microbiome association analyses using MaAsLin2 on shotgun metagenomic data from fecal samples obtained from two independent cohorts of patients with IBD (Mikinautes and Mucosa). MiR-21 and let-7b expression levels were modelled as continuous variables against bacterial taxonomic abundance, adjusting for clinical and demographic covariates. The identified associations were subsequently validated using complementary *in vitro* approaches. Targeted microbiological assays examined the effects of miR-21 and let-7b on selected bacterial phenotypes, whereas metatranscriptomic analyses characterized miRNA-induced changes in microbial functional pathways within a complex *in vitro*-cultured microbiota.

Results

Our analysis identified significant associations between let-7b expression and five bacterial species: *E. coli*, *R. gnavus*, *E. clostridioformis*, *F. umbilicata* and *B. massiliensis*, as well as an association between miR-21 expression and *A. inops*. Experimental validation of the direct effects of miR-21 and let-7b on the identified bacterial species is currently ongoing, with an initial focus on aerobic commensal *E. coli* K-12 and pathogenic *E. coli* LF82 strains, as well as the anaerobic species *R. gnavus*. Metatranscriptomic analyses revealed an overall increase in pathway activity following exposure to both miRNAs, with an overall trend of increased pathways activity rather than decreased. MiR-21 and let-7b both increased the activity of 73 shared pathways, predominantly involved in amino acid biosynthesis and the biosynthesis of cofactors, carriers and vitamins. In addition, let-7b increased the activity of 13 pathways, mainly related to energy metabolism; whereas miR-21 increased the activity of 50 additional pathways, the majority of which were involved in the biosynthesis of cofactors, carriers and vitamins.

Conclusion

Together, these findings provide further evidence for an interaction between miRNAs and the intestinal microbiota and suggest that miR-21 and let-7b may influence IBD-associated microbial communities through associations with specific bacterial taxa and direct effects on bacterial physiology and function. Further characterization of these interactions may provide new insights into the molecular mechanisms underlying host-microbiota crosstalk in IBD.

Modelling the Impact of Antiviral Therapy, Oral and Long-Acting Injectable Pre-Exposure Prophylaxis on HIV Incidence and Mortality among Sexually Active Men who have Sex with Men in metropolitan France (1972–2030)

Giang Ngoc Truong NGUYEN (1), Sylvie DEUFFIC (1), Liem Binh LUONG NGUYEN (1), Anthony COUSIEN (1), Amandine PEPIOT (1)

(1) PreViST, INSERM UMR 1137, IAME, France.

Introduction: HIV disproportionately affects men who have sex with men (MSM). Major prevention and treatment milestones, such as highly active antiretroviral therapy (HAART) and pre-exposure prophylaxis (PrEP), have transformed HIV into a manageable chronic condition. Yet, incidence and mortality remain far above the elimination targets, which include 90% reduction in HIV incidence and 90% AIDS-related mortality in 2030 relative to the 2010 baseline, according to the United Nations Programme on HIV/AIDS (UNAIDS).

Objectives: We aimed to determine when France will achieve the UNAIDS targets, and to model the effects of intervention scenarios, including scale-up or decline in viral suppression on antiretroviral therapy (ART) and effective PrEP coverage, and introduction of long-acting injectable PrEP (LA-PrEP), to identify under which conditions the 2030 targets could be met or, conversely, under which re-emergence could occur.

Methods: We developed a deterministic HIV transmission model formulated as a system of ordinary differential equations, spanning five epidemiological periods. The model was calibrated to empirical data from 1972 to 2025. We used Latin hypercube sampling (1,000,000 parameter sets) and selected best-fitting sets by minimizing the normalized root mean square error against multi-decade real-world data as calibration targets. We then projected HIV incidence for 2026-2030 under multiple ART and PrEP scale-up and decline scenarios, varying the proportion of LA-PrEP among all PrEP use.

Results: Under status-quo trajectories, the model projected that the 90% reduction target for HIV incidence would be reached in 2041, and for AIDS-related mortality in 2047. To achieve the 2030 target for HIV incidence, the model suggested that any one of the following (considered independently) would be required: increasing viral suppression coverage from the projected baseline of 89.3% in 2030 to 97.1%, increasing overall effective PrEP coverage from 11.5% to 27.3%, or increasing effective PrEP coverage among high-risk individuals from 33.2% to 70%. Additionally, in a scenario where we did not increase PrEP initiation, shifting 50% of PrEP use to LA-PrEP was projected to reduce annual HIV incidence by approximately 20%. On the contrary, if viral suppression coverage falls below 81% while overall effective PrEP coverage simultaneously declines below 3%, HIV incidence is projected to rebound to at least 30% above the 2010 baseline, indicating epidemic re-emergence.

Conclusion: Reaching the UNAIDS 2030 target in this population is unlikely under current trends and would require substantial scale-up of ART and/or PrEP, with LA-PrEP potentially providing meaningful benefits. Interruptions or reductions in these interventions would likely result in re-emergence. However, projections depend on modelling assumptions and data availability, requiring improvements in model structure and longitudinal surveillance. A cost-effectiveness analysis would provide a clearer estimate of the resources needed to achieve the target.

Optogenetic nanotool to investigate transcription factor activity

Morgane DE SOUSA (1), Yoann LALATONNE (1), Carine LE GOFF (2), Baptiste AMOUROUX (1)

(1) Team 4, Laboratory for Vascular Translational Science (LVTS), INSERM U1148, Université Sorbonne Paris Nord, Université Paris-Cité, France. (2) Team 2, Laboratory for Vascular Translational Science (LVTS), INSERM U1148, Université Sorbonne Paris Nord, Université Paris-Cité, France.

Transcription factors (TFs) are proteins that regulate gene expression in the nucleus. Their dysfunction can lead to the development of rare cardiovascular diseases. Understanding the role of dysfunctional TFs in rare cardiovascular diseases therefore represents a major challenge. However, currently available strategies often lack specificity and may induce pleiotropic effects, highlighting the need for novel approaches to investigate TF function with greater precision.

Among these emerging strategies, optogenetics enables precise spatiotemporal control of protein activity and facilitates functional analysis while minimizing unwanted pleiotropic effects. This approach relies on genetically engineered light-sensitive proteins, such as LOV2 [1], which can be activated on demand by blue light (450 nm). In contrast to conventional genetic modification strategies, the activity of these optogenetic biosensors is fully reversible, allowing dynamic control of protein function.

However, the limited tissue penetration of blue light restricts the use of optogenetics for in vivo applications. One promising solution is to combine LOV2 with upconversion nanoparticles (UCNPs), inorganic luminescent nanomaterials that can be activated by near-infrared (NIR) light (808 or 980 nm) and locally emit blue light. These nanoparticles can therefore act as nanoscale light sources, or “nanolamps”[2], enabling remote activation of light-sensitive proteins in deeper tissues.

The aim of this project is to develop a model nanohybrid combining UCNPs and LOV2 to investigate the function of dysfunctional transcription factors. This approach will provide a novel tool for studying TF activity with precise spatiotemporal control and reversibility, while potentially overcoming the limitations associated with conventional optogenetic approaches in in vivo models.

In this presentation, we will focus on two main aspects:

- (1) the characterization of UCNPs as nanoscale light sources for optogenetic activation; and
- (2) the cloning strategy used to generate a first candidate construct for the study of TFs.

References

1. Niopek D, Benzinger D, Roensch J, Draebing T, Wehler P, Eils R, et al. Engineering light-inducible nuclear localization signals for precise spatiotemporal control of protein dynamics in living cells. *Nat Commun.* 14 juill 2014;5(1):4404. doi:10.1038/ncomms5404
2. Hudson DE, Hudson DO, Wininger JM, Richardson BD. Penetration of Laser Light at 808 and 980 nm in Bovine Tissue Samples. *Photomed Laser Surg.* avr 2013;31(4):163-8. doi:10.1089/pho.2012.3284

Role of hormonal and microbiome modulation on gastric emptying and metabolic response after endoscopic sleeve gastroplasty (HOVIGAMI-Endosleeve): a prospective comparative study

Maxime AMOYEL (1), Maude LE GALL (1), Lara RIBEIRO (1), Abdellah HEDJOUDE (1), Tigran POGHOSYAN (1)

(1) Equipe PIMS - Centre de recherche sur l'inflammation - Université de Paris Cité - France.

Introduction and motivation

Obesity is a growing public health problem with major metabolic consequences. Endoscopic sleeve gastroplasty (ESG) offers a less invasive alternative to bariatric surgery, with total body weight loss of roughly 13 to 16 percent at one year. Yet its underlying mechanisms remain poorly understood. Weight loss after bariatric interventions does not rest on mechanical gastric restriction alone: it involves gastric emptying dynamics, digestive hormone secretion, and possibly the gastric microbiome. While surgical sleeve gastrectomy accelerates gastric emptying, preliminary data suggest ESG slows it. Our team has shown in a prospective cohort of 23 patients that gastric emptying at 3 months and gastric volume at 1 month predict weight loss after ESG. However, the hormonal and microbial determinants of this response are still uncharacterized.

Methods

HOVIGAMI-Endosleeve is a prospective comparative study enrolling 40 patients with moderate to severe obesity (BMI 30 to 40 kg/m²), assigned to ESG or surgical sleeve gastrectomy according to standard multidisciplinary practice, with an 18-month inclusion period and 12-month follow-up. The primary endpoint is the change in solid-phase gastric half-emptying time ($T^{1/2}$), measured by gastric scintigraphy, from baseline to M1, M3, M6, and M12, compared between groups. Secondary assessments, performed at the same timepoints, include gastric volumetry by CT (gastroscanner), post-prandial digestive hormone kinetics (GLP-1, PYY, GIP, ghrelin, insulin, glucagon, C-peptide; AUC 0 to 180 min) after a standardized test meal, gastric microbiome analysis, anthropometric and metabolic parameters, and safety. Longitudinal multivariate models will identify the physiological determinants of the weight response.

Expected results

We expect to confirm that ESG slows gastric emptying, in contrast to laparoscopic sleeve, and to characterize the associated hormonal and microbial adaptations. Integrating gastric emptying, enteroendocrine signaling, microbiome, and gastric volume within a single longitudinal cohort should clarify the mechanisms of weight loss and help explain the heterogeneity of individual responses.

Significance and impact

By dissecting the physiological basis of ESG and laparoscopic sleeve, HOVIGAMI-Endosleeve could support better patient selection, personalized bariatric strategies, and the identification of predictive biomarkers, opening the way to combined endoscopic and pharmacological approaches for obesity.

Nationwide digital surveillance reveals resurgence of pertussis through real-time decision support systems

Pauline CHAUMET (1), Tristan DELORY (2), Josselin LE BEL (3), Cédric LAOUEANAN (4)

(1) Team MOCLID, Université Paris Cité, INSERM IAME UMR 1137, Paris, France, (2) Centre Hospitalier Annecy Genevois, Epagny Metz-Tessy, France ; Sorbonne Université, INSERM, Institut Pierre Louis d'Épidémiologie et de Santé Publique, Paris, France. (3) Université Paris Cité, INSERM IAME UMR 1137, Paris, France ; Département de médecine générale, Université Paris Cité, France. (4) Université Paris Cité, INSERM IAME UMR 1137, Paris, France; AP-HP Nord, Hôpital Bichat, Département d'Epidémiologie Biostatistique et Recherche Clinique, Paris, France.

Background and objective: The global burden of pertussis remains difficult to assess, as it follows a cyclical pattern with epidemic peaks every 3 to 5 years. In France, as in many Western countries, the incidence has remained low since the 1970s due to widespread vaccination. However, 2024 was marked by a major resurgence. The objective of this study is to objectify the ability of a nationwide computerised decision support system (CDSS) to characterise pertussis resurgence in primary care.

In this study, we investigated whether the patterns of use of the Antibioclie CDSS could detect signals of a pertussis resurgence using traditional surveillance networks as reference point. We used the CDSS requests intended to guide antibiotic prescribing by general practitioners for a range of more than 40 infectious diseases (including pertussis), to explore the population-level incidence of pertussis in primary care consultations in France between 2018 and 2024.

Material and Methods: We calculated annual and quarterly incidence rate ratio (RR) for 2024 and 2025 compared to 2018-2019. Stratified analyses were performed by age-group (adult or children), chronic kidney failure and pregnancy status. To quantify excess incidence in 2024, we applied a seasonal autoregressive integrated moving average (SARIMA) model trained on 2018–2023 data and projected 2024. Finally, we assessed the spatial distribution of pertussis at departmental-level, stratifying incidence for demographics and medical density, and compared findings with the previous year and the national vaccination data coverage (SNDS).

Results: The CDSS request analysis identified the outbreak four weeks prior to the national alert. Weekly incidence rose from ~ 1 per 100,000 inhabitant the prior years to 20 per 100,000 inhabitants at the peak (three weeks duration) in July 2024. The annual risk ratio for 2024 versus 2018 was 8.87 (95% CI, 8.79–8.95), with a marked increase in the third quarter with an 18.95 risk ratio (95% CI, 18.62-19.28). Requests increased particularly for children under 15 years old. The SARIMA model estimated a requests relative excess incidence of 1 348 % in 2024. Finally, spatial analysis revealed substantial heterogeneity and clusters of cases, with discrepancies between children and pregnant women, partly unexplained by variations in vaccination coverage.

Conclusion: CDSS data provide valuable real-time insights into pertussis epidemiology. Our findings confirm the exceptional resurgence of pertussis in France in 2024, identify vulnerable groups, and highlight heterogeneous departmental dynamics unexplained by vaccination gaps. Our findings emphasise the importance of real-time digital health tools in epidemiological surveillance.

Posters

ABSTRACTS

A Multimetric Framework for Standardized Assessment of Mass Spectrometry Imaging Protocols: Hepatocellular Carcinoma as a Proof of Concept

Ghalia BEN GHEDIFA (1), Nathalie COLNOT, Miguel ALBUQUERQUE, H     CAZIER, Val     PARADIS

(1) INSERM U1149.

Introduction

Mass spectrometry imaging (MSI) has demonstrated strong performance for FFPE (formalin fixed, paraffin-embedded) tissues analysis, providing molecular segmentations that generally align with pathologists' annotations. Achieving repeatable and reproducible MALDI-MSI (Matrix - Assisted Laser Desorption - Ionisation) data in heterogenous tissues - especially primary liver tumours (such as hepatocellular carcinoma (HCC)) - often require extensive sample preparation optimisation. Quantitative criteria are therefore required to objectively benchmark workflows. Here, we propose a multi-metric framework to assess optimisation of an in-situ bottom-up proteomic MALDI-MSI workflow on HCC tissues.

Methods

3  m FFPE sections of HCC tissue were mounted on indium tin oxide (ITO) slides for peptidomic analysis using MALDI-TOF (time-of-flight) mass spectrometry (Rapiflex, Bruker). After preheating, two deparaffinisation/rehydration protocols and 5 antigen retrievals buffers (water, citraconic acid, citrate, Tris-HCl and Tris-EDTA) were assessed prior tryptic digestion. Samples then underwent trypsin spray, 3H digestion at 50   C and HCCA matrix deposition using the M3+ sprayer (HTX Technologies). Data were TIC-normalised, segmented and exported as .imzML/.ibd in SciLS Lab (Bruker) for R analysis. 6 protocols were compared using spectral and spatial reproducibility metrics as m/z overlap, inter-replicate intensity variability and segmentation stability, based on shared m/z features 0.25 Da and intensity threshold > 3.

Main results

HCC FFPE tissue analyzed in triplicate reveal clear protocol dependent effects of the antigen retrieval using our multi-metric framework. While the number of detected features remained broadly comparable across conditions (~450  465 up to 569 peak detected) major differences emerged in signal intensity. The Water-based retrieval protocol has shown a balanced compromise between metrics, with ~ 450-465 features per pixel, a TIC median at 560, good reproducibility (CV ~16  18%) and spatial coherence (~0.63  0.69). Ongoing work will extend our methodology to more samples and experimental optimisation.

Deciphering the catalytic mechanisms and substrate specificity of *Klebsiella pneumoniae* carbapenemase using site-directed mutagenesis and deep mutational scanning

Noémie RYBAK (1)(2), Yasmine BENHADID-BRAHMI (1)(2), Léonie LEGENDRE (1)(2), Stéphane BONACORSI (1)(2), André BIRGY (1)(2)

(1) Laboratoire de microbiologie, Hôpital Robert-Debré, AP-HP, Paris, France. (2) Université Paris Cité, USPN, IAME, UMR 1137, INSERM, Paris, France.

Carbapenem resistance in Enterobacterales is a WHO critical priority. KPC, a class A serine- β -lactamase, displays remarkable evolutionary plasticity that has driven resistance to ceftazidime/avibactam (CAZ-AVI), the current standard of care. While mutational hotspots in loops surrounding the active site are well characterized, the precise contribution of individual residues to the catalytic mechanism and substrate specificity of KPC-2 remains incompletely defined. This work aimed to define the individual contribution of key catalytic and structural residues of KPC-2.

Materials and methods

Alanine scanning of 19 catalytic and structural residues of KPC-2 was performed. Phenotypic resistance profiles were evaluated against a broad panel of β -lactams. For each molecule, the minimal inhibitory concentration shift relative to ancestral KPC-2 scored the residue's contribution to its hydrolysis. In parallel, blaKPC-2 was cloned into pSKUNK and uracil-containing single-stranded DNA produced in *E. coli* CJ236, the template for deep mutational scanning (PFunkel).

Results

Loss of resistance to a given β -lactam identified the residue as required for its hydrolysis, ranking the 19 substitutions into five functional categories. The catalytic tetrad (S70, K73, S130, E166) proved essential for β -lactamase activity, with the notable finding that E166A retained a ceftazidime MIC of 8 mg/L, suggesting acyl-enzyme trapping analogous to certain clinical variants. Residues N132, N170, and T235 were identified as critical for carbapenem resistance, while W105, R164, and R220 function[MD1.1] as modulators. Substitution-tolerant residues (S171, V240, H274) were also identified, with effects manifesting primarily in double mutant contexts. Notably, D179A was the sole substitution conferring a ceftazidimase profile with CAZ-AVI resistance[MD2.1].

Conclusion

These results refine the structure-activity relationships of KPC-2 by distinguishing residues that are strictly essential from those that modulate substrate specificity, and identify D179 as a pivotal position in the transition toward a ceftazidimase phenotype. The single-stranded DNA platform developed here opens the way to mapping the full mutational landscape accessible to KPC-2 under antibiotic selective pressure.

The resistome of *Enterococcus*

Mattéo LEGUEN (1), Vincent CATTOIR (2), Etienne RUPPE (1)

(1) EVRest, IAME, Inserm / AP-HP, Hôpital Bichat-Claude Bernard. (2) Inserm U1230 BRM, University of Rennes / National Reference Center for Enterococci, University Hospital of Rennes / Department of Bacteriology, University Hospital of Rennes.

Background

Enterococcus species are commensal inhabitants of the human and animal gastrointestinal microbiota, but are also major opportunistic pathogens in healthcare settings, responsible for urinary tract, intra-abdominal and bloodstream infections, and endocarditis. Their intrinsic and acquired resistance to multiple antibiotics, including high-level resistance to ampicillin and vancomycin, considerably limits therapeutic options, particularly for vancomycin-resistant enterococci (VRE). Unlike staphylococci, beta-lactamase production has historically been considered rare in *Enterococcus*. High-throughput sequencing and public genomic databases such as EnteroBase now offer the opportunity to reassess antibiotic resistance gene (ARG) distribution at unprecedented scale, combining tens of thousands of assembled genomes with rich metadata. However, this exploitation raises methodological challenges, including cross-contamination, variable assembly quality, and inconsistently curated metadata, particularly problematic when investigating rare ARGs. This study aims to characterize ARG diversity and distribution in *Enterococcus*, focusing on *Enterococcus faecium* and *Enterococcus faecalis*, using the EnteroBase dataset.

Materials and methods

A total of 54,037 genomes were downloaded from EnteroBase via API access (March 2026), with associated metadata. Rigorous quality control was applied using CheckM2 (completeness >90%, contamination <5%, N50 >30,000), genome size consistency with NCBI reference ranges, and concordance between MLST-predicted and metadata-associated species, retaining 45,993 genomes. ARG detection was performed via BLAST against the ResFinder database (identity/coverage thresholds ≥80%), with deduplication based on the highest bitscore. Species-specific MLST schemes were used for *E. faecium* and *E. faecalis*. Mann-Whitney and Chi² tests were used for comparisons (significance threshold $p < 0.05$).

Results

E. faecium (75.0%) and *E. faecalis* (19.7%) were the most prevalent species, followed by *E. hirae*, *E. gallinarum*, and *E. durans* (each <2.5%). Isolates mainly originated from human sources (56.8%) and high-income countries (79.8%), with the US, Denmark, and Australia predominating. A total of 238,075 ARGs (139 unique) were detected. *E. faecium* carried significantly more ARGs than *E. faecalis* (6.03 vs 2.78 per genome; $p < 10^{-6}$), with marked prevalence of vanHAX and vanHBX operons — showing strong geographic (vanHAX predominant in Ireland/US/UK, vanHBX in Germany/Switzerland) and clonal structuring (vanHBX nearly fixed in ST796) — and a broad aminoglycoside resistance repertoire, more prevalent in human than livestock or poultry isolates. Conversely, *E. faecalis* was characterized by high prevalence of tet(L) and tet(M), the latter showing pronounced ST-dependent heterogeneity (absent in ST6/ST40, nearly fixed in ST16/ST179), while tet(L) varied markedly by geography (highest in China and Belgium). Only one *E. faecalis* genome (China, novel ST) carried a beta-lactamase gene (blaTEM-116), after excluding 46 other candidates deemed unreliable following quality filtering.

Discussion

This large-scale analysis reveals distinct resistomes between *E. faecium* and *E. faecalis*, strongly structured by sequence type, country, and isolation source, likely reflecting clonal expansion of resistant lineages and circulation of specific mobile genetic elements. Rigorous quality control proved essential for interpreting rare ARGs, ruling out broad beta-lactamase dissemination in *Enterococcus* and underscoring the necessity of stringent contamination and assembly-quality filters when exploiting large public genomic collections

A high-fat diet is insufficient to drive pancreatic intraepithelial neoplasia from acinar-to-ductal metaplasia

Louis MARSTRAND-DAUCE (1), Louis PITON (1), Anaïs CHASSAC (1)(2), Anne COUVELARD (1)(2), Vinciane REBOURS (1)(3), Cécile HAUMAITRE (1)

(1) INSERM, U1149 / Inflammation Research Center (CRI), Bichat Medical Faculty, Université Paris Cité, Paris, France. (2) Université Paris Cité, Department of Pathology, Bichat Hospital, APHP, Paris, France. (3) Université Paris Cité, Pancreatology and Digestive Oncology Department, Beaujon Hospital, APHP, Clichy, France.

Aim

Activation of KRAS is a key initiating event in pancreatic ductal adenocarcinoma (PDAC) and drives pancreatic intraepithelial neoplasia (PanIN). Obesity and high-fat diet (HFD) are associated with increased PDAC risk. HFD has been shown to accelerate pancreatic tumorigenesis in KRAS-driven mouse models. However, whether HFD-induced metabolic stress can initiate PanIN lesions independently of oncogenic KRAS remains unclear.

Methods

We assessed the effect of HFD in a murine model with a permissive pancreatic microenvironment characterized by spontaneous acinar-to-ductal metaplasia (ADM) and intrapancreatic adipocyte infiltration (Sox9-CreER; Hnf1b^{fl/fl}, hereafter HC mice). Mice were fed either a normal diet (ND) or a HFD (60% fat). Metabolic parameters including body weight, glucose tolerance, cholesterol and triglyceride levels were monitored. Pancreatic tissues were analyzed by histopathology, and morphometric analyses evaluated ADM and PanIN lesions. Immunohistochemistry assessed epithelial proliferation and inflammatory markers.

Results

HFD induced metabolic alterations including increased body and pancreatic weight, glucose intolerance, and elevated circulating lipids. Despite a pancreatic microenvironment permissive for metaplastic changes, HFD failed to induce PanIN lesions in HC mice. Ki67 immunostaining revealed significantly increased epithelial proliferation in HFD-fed mice compared with ND controls. Quantitative analyses showed no difference in ADM extent between HC-ND and HC-HFD after normalization to pancreatic weight.

Conclusion

Although HFD induced metabolic stress and increased epithelial proliferation, it was insufficient to promote PanIN formation. These findings indicate that metabolic stress alone cannot initiate pancreatic carcinogenesis in the absence of oncogenic KRAS, supporting a model in which HFD acts as a tumor-promoting rather than tumor-initiating factor.

Etude du syndrome de Myhre : Caractérisation du phénotype cardiaque et pulmonaire du modèle murin Smad4 Ile500/+

Angélique BIBIMBOU (1), Morgane DE SOUSA (1), Hayatte MAIBECHE (1), Marie-Nathalie LEBEL (1), Zakaria MOUGIN (1), Carine LE GOFF (1)

(1) Paris Cité Université and Sorbonne Paris Nord University, INSERM UMRS1148 Team 2, Laboratory for Vascular Translational Science, Paris.

Le syndrome de Myhre (SM) fait partie des dysplasies acroméliques, des maladies génétiques rares. Il se caractérise par une petite taille, un épaissement cutané, une perte auditive et des anomalies cardiovasculaires observées chez environ 70 % des patients ainsi que des atteintes pulmonaires. Plusieurs variants associés au SM ont été identifiés dans le gène SMAD4, principalement au niveau des résidus Arg496 et Ile500. En tant que co-médiateur des voies de signalisation TGF β et BMP, SMAD4 est essentiel au développement normal et au maintien de l'homéostasie de l'ensemble de l'organisme.

Nous avons généré un nouveau modèle murin en introduisant la mutation Ile500 dans le gène SMAD4. Les premières études ont montré que ce nouveau modèle Smad4I500V/+ est viable et présente un retard statural associé à des anomalies du cartilage de croissance et de la différenciation des chondrocytes. Les souris homozygotes Smad4I500V/I500V présentent un phénotype plus sévère, avec une létalité périnatale.

Afin de comprendre cette létalité et le syndrome de Myhre, l'objectif de ce travail est de caractériser les phénotypes cardiovasculaire et pulmonaire associés au MS et d'étudier le mécanisme physiopathologique lié aux mutations SMAD4 dans ce syndrome.

Une analyse des caractéristiques morphologiques des poumons du modèle murin Smad4I500V/+ a été réalisée. Les poumons des souris mutées montrent une diminution de l'aire alvéolaire de environ 30% chez les HT et 35% chez les Ho ainsi qu'une augmentation des septas de 12% chez les HT à P1. En revanche, on observe chez les Ho une diminution des septa d'environ 8%.

De plus, les cœurs mutants sont plus petits que les cœurs des souris sauvages avec des parois plus épaisses. De façon intéressante, les cœurs des souris homozygotes présentent un phénotype plus sévère avec des défauts de septation.

Afin de comprendre ses défauts de développement du cœur et grâce au séquençage RNA-seq, nous avons établi le profil transcriptomique des cœurs mutants et identifié que la majorité des ARNm exprimés différemment sont associés aux voies du cil primaire suggérant l'implication de cette structure dans le phénotype cardiaque du SM. Nous avons également constaté des défauts d'expression de gènes de la matrice extracellulaire.

Ces résultats suggèrent un rôle du cil primaire dans le mécanisme physiopathologique ainsi que l'implication de SMAD4 dans le phénotype pulmonaire notamment avec une altération morphologique. L'implication de la matrice extracellulaire dans la physiopathologie du syndrome de Myhre fait écho à celle d'autres dysplasies acroméliques associées à des mutations dans FBN1 et ADAMTSL2, qui codent tous deux pour des protéines de la matrice extracellulaire. Ces données apportent ainsi de nouvelles perspectives dans la compréhension des mécanismes physiopathologique du SM.

Multi-response extension and computational optimisation of a framework coupling non-linear mixed-effects modeling with whole-body PBPK models

Lucas BODELLE (1), Marc LAVIELLE (2), David MARCHIONNI (3), Laurent NGUYEN (4), Emmanuelle COMETS (1), Donato TEUTONICO (4)

(1) Université Paris Cité and Université Sorbonne Paris Nord, Inserm, IAME, F-75018 Paris, France. (2) Inria, 91400 Saclay France and Ecole Polytechnique, 91120 Palaiseau France. (3) Sanofi, Translational Medicine Unit, 34000 Montpellier France. (4) Sanofi, Translational Medicine Unit, 94400 Vitry sur Seine France.

Background

Quantifying variability sources in drug concentrations through non-linear mixed effect models (NLMEM) has proven instrumental in drug development [1] and can be performed in commercial (Nonmem, Monolix) or open-source software like the R package saemix [2]. More complex Physiologically Based Pharmacokinetic (PBPK) models involve numerous processes and combine data from multiple sources [3], usually through specialised software such as PKSim and the R package opsuite from the Open System Pharmacology (OSP) Suite [4]. However, estimation approaches are mostly limited to individual regression with less flexibility for exploring interindividual variabilities. Recently, the saemixPBPK package coupling saemix with PBPK/QSP models developed with OSP Suite was created to bridge PBPK modelling and pharmacometrics approaches [5]. This framework allows estimation of population parameters and their inter-individual variability (IIV) for mechanistic parameters in PBPK models.

Objectives

In this work, we estimate physiological parameters with IIV in a whole-body PBPK model with multiple responses (drug and metabolites). We extend the saemix package to handle multiple responses, also computationally improving the saemixPBPK grid approximation algorithm.

Methods

A PBPK model was developed for drug SAR (Sanofi) with clearance partly attributed to CYP3A4 metabolism. While the model integrated in vitro measurements of CYP3A4 metabolizing fraction (Fm), it overpredicted this parameter based on data from a drug-drug interaction (DDI) study with itraconazole (n=8 subjects). In a prior analysis, two feasible scenarios were proposed to estimate CYP3A4 Fm using saemixPBPK[6], with long computational times. We first enhanced saemixPBPK computational efficiency through parallel computation capabilities, speed optimization of the grid approximation algorithm via C++ code rewriting, and data structure refinement for improved memory usage. We also extended saemix to handle multiple responses for integration with saemixPBPK, to simultaneously model itraconazole and hydroxy-itraconazole. We were then able to simultaneously estimate CYP3A4 abundance and itraconazole catalytic constant with IIV. Subsequently, individual CYP3A4 fraction metabolized (Fm) was estimated using data from the interaction arm.

Results

The estimated individual fraction metabolized values using individual plasma concentrations (itraconazole and its metabolite) with saemixPBPK were consistent with classical PBPK estimation approaches but additionally provided estimates of variabilities and uncertainty. The computational optimisation implemented in saemixPBPK resulted in a 300% speed improvement and 98% reduction in memory usage.

Conclusion

This methodology bridges traditional population pharmacokinetic modeling with mechanistic PBPK modeling through open-source software development, enabling simultaneous estimation of IIV and population parameters across multiple responses from individual concentration data

Deciphering the role of the recently identified eefRABCD efflux pump in E.coli.

Rime CHEBBO, Evrest, CB3S, USPN.

Antibiotic resistance has become a major public health challenge. In parallel, pathogenic clones, characterized by a large number of virulence associated genes, have emerged and spread in the community over the last decades. While antibiotic resistance and virulence have traditionally been considered as distinct traits, increasing evidence suggests that some bacterial mechanisms may contribute to both. Among these, Resistance-Nodulation-Division (RND) efflux pumps are well known for mediating multidrug resistance through the active export of a wide range of antimicrobial compounds. Recent studies suggest that these systems may also influence bacterial adaptation, host colonization and virulence. Although the major RND efflux pumps of *E. coli* have been extensively characterized, some of them remain poorly understood with respect to their distribution, physiological function and contribution to bacterial adaptation and virulence.

First, we investigated the distribution of RND efflux pumps across *E. coli* phylogroups, using genomic sequences of 914 clinical and 425 commensal isolates. While several pumps were largely distributed across the both collections, the eefRABCD efflux pump displayed a distinct distribution. This pump was first described in *Enterobacter* species and recently identified in *E. coli*. Recent published results found this pump to be highly prevalent across phylogroups B2, D, E, F and G which encompass most classical extraintestinal pathogenic *Escherichia coli* lineages, while it was absent from phylogroups A, B1 and C. Virulence (severity of infection) and pathogenicity (capacity to cause infection in humans) were previously defined by other groups for many strains in a mouse sepsis model and using genome wide association analyses respectively. While we did not find any association between the presence of this pump and virulence, we found an association with pathogenicity. Among all RND systems, eefRABCD also showed the strongest association with about 90 virulence-associated genes, whereas no significant association with antimicrobial resistance genes was observed.

Next, to investigate the functional role of this pump, we deleted the entire eefRABCD operon in the uropathogenic UTI89 strain. Our preliminary results showed no significant differences in bacterial growth or antimicrobial susceptibility in Muller Hinton media compared with the wild-type strain, suggesting that this efflux system may not primarily contribute to antibiotic resistance under the conditions tested. Accordingly, our future work will aim to assess the contribution of this pump to adaptation, antibiotic response in diverse environmental contexts including in vitro competition assays and in vivo mouse gut colonization and infection models. Additionally, representative isolates will be selected for molecular characterization followed by complementary Nuclear Magnetic Resonance (NMR)-based metabolomics to investigate the metabolic adaptations and energetic costs associated with efflux activity under different conditions, providing mechanistic insights into its physiological role and association with particular phenotypes.

Development of a pulmonary fibrosis model in immune-humanized BRGSF mice

Nathan HENNION (1), Zackarya KARAMANLIAN (1), Fabiane SÔNEGO (2), Philippe DUTHEIL DE LA ROCHERE (2), Kader THIAM (2), Bruno CRESTANI (3), Georges DOUMET HELOU (1)

(1) Junior Group Helou, Unité INSERM UMR1149, Centre de Recherche sur l'Inflammation, Paris, France. (2) GenOway, Lyon, France. (3) Team Mailleux/Crestani, Unité INSERM UMR1149, Centre de Recherche sur l'Inflammation, Paris, France.

Introduction

Idiopathic pulmonary fibrosis (IPF) is an interstitial lung disease characterized by the irreversible accumulation of extracellular matrix (ECM) leading to the destruction of lung tissue. Current antifibrotic treatments slow disease progression but do not cure it. The lack of curative treatments is partly due to the limited availability of models capable of recapitulating the pathophysiological mechanisms underlying IPF. Recent transcriptomic approaches have shown that the immune system actively contributes to the fibrotic microenvironment. To better recapitulate the immune mechanisms involved in IPF, we proposed to induce pulmonary fibrosis (PF) in the immunodeficient BRGSF mouse reconstituted with human immune cells (HIS), which we recently characterized in terms of lung immune composition.

Methods

To develop BRGSF-HIS mice, sublethally irradiated BRGSF newborn mice were reconstituted by intrahepatic injection of human hematopoietic cells. In vitro, human monocyte-derived macrophages were exposed to a fibrotic cocktail (FC) and their fibrotic activity was assessed by flow cytometry and RT-qPCR. Culture supernatants were collected for cytokine assays. In vivo, mice were intranasally treated with bleomycin (BLM) to induce and study PF development on day 14. Bronchoalveolar lavage and lungs were collected to characterize the activity of human immune cells (hCD45+) by spectral cytometry and to quantify relevant cytokines. PF was assessed by histology, RT-qPCR, and Western blotting.

Results

Our results showed that cultured macrophages exposed to the FC upregulate the expression of pro-fibrotic genes such as CCL18, IL10 and ARG1 and the production of IL-6, IL-1RA and CCL17. Moreover, BLM administration induces PF, characterized by the accumulation of collagen I α and fibronectin 1. Histological analyses revealed an accumulation of ECM in the lung parenchyma. Spectral cytometry showed increased recruitment of diverse myeloid and lymphoid hCD45 + cells to the lung, with T lymphocytes being the most abundant population. Also, the levels of human CCL17 and IL-12p40, were increased in the lung.

Conclusion

Altogether, this study reveal that humanized BRGSF-HIS mice successfully develop PF in response to BLM, implicating human immune responses. Although our findings represent a promising first step toward developing a relevant PF model with human immunity, further optimization is needed to enhance myeloid cell recruitment and more faithfully recapitulate the immune features of IPF.

Epigenetic control of HDAC3: From Pancreatic Regeneration to Tumor Initiation

Linkang MENG (1), Anaïs CHASSAC (1), Marie DUMONT (1), Louis MARSTRAND-DAUCE (1), Diane LORENZO (1), Pascal NICOLE (1), Valérie GRATIO (1), Anne COUVELARD (1)

(1) Equipe Haumaitre / Cros - "PANcreatic Tumorigenesis Heterogeneity (PATH)" UMR 1149 Inserm - Centre de Recherche sur l'Inflammation (CRI).

Introduction – Pancreatic ductal adenocarcinoma (PDAC) is one of the most lethal cancers. PDAC arises from exocrine pancreatic cells, including acinar cells, and is often preceded by chronic inflammation, such as pancreatitis. Pancreatitis disrupts pancreatic homeostasis and can trigger acinar-to-ductal metaplasia (ADM), a process that may progress to pancreatic intraepithelial neoplasia (PanIN), a key precursor to PDAC. Understanding the molecular mechanisms driving these lesions is crucial for identifying new therapeutic targets. This study investigates the role of Histone Deacetylase 3 (HDAC3) in pancreatic regeneration and cancer initiation.

Methods – To assess the role of HDAC3, we utilized genetically engineered mouse models with conditional inactivation of *Hdac3* specifically in pancreatic acinar cells (HP mice). Mice were subjected to models of acute and chronic pancreatitis to evaluate the impact of HDAC3 loss on pancreatic regeneration and lesion formation. Given that ADM can lead to PanIN formation, we further examined the role of HDAC3 in the initiation and progression of these pre-neoplastic lesions in a tumor-prone environment. This was achieved by activating KRASG12D in models with and without inflammation induced by cerulein (KHP mice). *Hdac3*-deficient acinar cells were isolated and cultured in a growth-factor-free and TGF α -free matrix, conditions that do not promote ADM. These cultures were analyzed to assess their intrinsic capacity to undergo ADM. Additionally, RNA sequencing (RNA-Seq) was performed on isolated pancreatic acinar cells from HP, KHP and respective controls H and KP (Ptf1a-CreER;LSL-KRASG12D) to identify transcriptional and epigenetic alterations associated with *Hdac3* inactivation. Selected differentially expressed genes were further assessed by RT-qPCR.

Results – Our findings demonstrate that HDAC3 plays a critical role in pancreatic regeneration following injury. *Hdac3* deficiency led to increased ADM lesions and heightened immune infiltration, worsening pancreatic damage after acute and chronic pancreatitis. Furthermore, the absence of HDAC3 promoted PanIN formation showing that *Hdac3* deficiency facilitates early neoplastic transformation mediated by KRASG12D. HDAC3 loss further exacerbated PanIN formation when combined with both KRASG12D activation and inflammation. Ex vivo, *Hdac3*-deficient acinar cells exhibited a significantly higher and more rapid conversion to ADM compared to control cells, highlighting their intrinsic propensity to undergo this transformation. RNASeq analysis of acinar cells from H, HP, KP and KHP mice revealed significant dysregulation in metabolic pathways, transcriptional regulation, epigenetic modifications in the absence of HDAC3. Pathway enrichment analysis highlighted AKT-mTOR pathway. Differential expression analysis identified increased Akt and decreased Pten expression in HP cells compared with controls and these changes were confirmed by RT-qPCR.

Conclusion – These results establish *Hdac3* as a key regulator of acinar cell regeneration and neoplasia initiation. Its loss promotes the transition from pancreatitis-associated injury to precancerous lesions, underscoring its potential as a biomarker for early pancreatic cancer detection. In addition, it highlights that non-specific HDAC inhibitors could be detrimental if they also target HDAC3. Further investigation into HDAC3-associated pathways could identify novel therapeutic targets for pancreatitis and PDAC prevention.

Magneto-Plasmonic Polysaccharide Particle-Assisted Thrombolysis

Adeline DOS SANTOS, Team 4, LVTS, USPN.

Thrombosis is responsible for most strokes and heart attacks, which are the two leading causes of death worldwide. Reperfusion is commonly performed with thrombolytics such as recombinant tissue plasminogen activator (rt-PA) but these treatment present numerous limitations (severe adverse effects, few eligible patients, low recanalization rate). With the objective of designing a safer and efficient treatment, we designed hybrid particles for a targeted physical action on the thrombus. This non-pharmaceutical thrombolysis combines thermal and mechanical action, using light and magnetic stimulation, and would be targeted to the occluded vessel by an external permanent magnet.

To achieve this strategy, we incorporate iron oxide nanoparticles that have great photothermal and magnetic properties in polysaccharide microbeads that are stable and intravenously injectable (Patent – INSERM).

We confirmed the hybrid structure by electron microscopy and FT-IR spectroscopy. Through dynamic light scattering (DLS) and electrophoretic light scattering (ELS), we studied the time stability in size and surface charge of these new microsparticles and characterized them also in saline solution. Magnetization measurements and particle counting were performed to compute the magnetic moment of these hybrid particles. Also, their response to magnetic stimulation was confirmed. This was completed with photothermal tests to estimate their ability to increase temperature under near-infrared laser irradiation. In an in vitro flow model (platelet binding assay), these particles could be retained by a magnet while flowing in venous and arterial conditions, proving their ability to magnetically target the thrombus. Finally, the effect of different levels of hyperthermia was evaluated in vitro on blood clots.

Based on these promising results, a second generation of beads with a modified composition is currently being developed to further optimize their photothermal properties and improve their thrombolytic efficiency.

Le tirzépatide remodèle l'identité transcriptomique des cellules caliciformes dans le jéjunum d'un modèle de rat de diabète de type 2

Mélanie GARMON, Equipe PIMS, Centre de Recherche sur l'Inflammation.

Introduction et but de l'étude

Le diabète de type 2 (DT2) est caractérisé par une homéostasie glucidique altérée résultant d'une insulino-résistance et d'un dysfonctionnement des cellules β . Le tirzépatide, double agoniste des récepteurs GIP et GLP-1, améliore le contrôle glycémique des patients, mais ses mécanismes d'action intestinaux restent mal compris. Nous avons étudié les effets métaboliques et intestinaux d'un traitement chronique par tirzépatide chez le rat Goto-Kakizaki (GK), modèle murin de DT2 spontané non obèse.

Matériel et méthodes

Deux cohortes indépendantes de rats mâles GK ont été utilisées. Dans la première, les rats ont reçu des injections sous-cutanées quotidiennes de tirzépatide ($n=7$) ou de véhicule ($n=7$) pendant 8 semaines, avec suivi quotidien du poids et de la prise alimentaire. Un test de tolérance orale au glucose et aux lipides (OLGTT) et un test de sensibilité à l'insuline (ITT) ont été réalisés après 3 semaines, avec calcul des AUC pour chaque animal. Les résultats sont exprimés en moyenne $\pm$ sem. Les comparaisons temporelles entre groupes ont été réalisées par Two-Way ANOVA repeated measures (post-hoc de Šidák), et les AUC par test de Mann-Whitney. Dans la seconde cohorte, les rats ont été traités pendant 3 semaines ($n=3$ /groupe) puis sacrifiés pour isolement des cellules épithéliales jéjunales et scRNA-seq (Parse Biosciences). Parmi 25 805 cellules intestinales séquencées, 2 981 cellules caliciformes ont été identifiées et soumises à l'analyse transcriptomique différentielle (TrailMaker).

Résultats et analyse statistique

La tolérance au glucose est améliorée chez les rats traités par tirzépatide, avec une diminution de l'AUC de l'OLGTT (-18,4%, $p=0,007$) et une glycémie réduite de T10 à T60 ($p<0,05$), sans modification de la glycémie à jeun ni de la sensibilité à l'insuline. Le poids corporel a diminué durablement (-13% à 8 semaines, $p<0,0003$) malgré une reprise rapide de la prise alimentaire. L'analyse scRNA-seq montre que le tirzépatide induit un enrichissement en cellules caliciformes différenciées par rapport au phénotype prolifératif, avec 635 gènes sur-exprimés et 1 106 sous-exprimés ($p\text{-adj}<0,05$). Les gènes sur-exprimés sont enrichis en voies de biosynthèse du cholestérol ($p\text{-adj}=1,33\times 10^{-8}$), d'intégrité des jonctions serrées ($p\text{-adj}=0,004$) et de translocation de GLUT4 ($p\text{-adj}=0,007$). Les gènes sous-exprimés sont enrichis en voies du cycle cellulaire ($p\text{-adj}=3,25\times 10^{-18}$), de réplication de l'ADN ($p\text{-adj}=3,26\times 10^{-12}$) et de glycolyse ($p\text{-adj}=0,004$). L'absence d'expression des récepteurs aux incrétines dans les cellules caliciformes suggère un mécanisme d'action indirect.

Conclusion

Le tirzépatide améliore la tolérance au glucose et induit une perte de poids durable chez le rat GK malgré une reprise rapide de la prise alimentaire, suggérant des adaptations métaboliques indépendantes de la seule restriction calorique. Au niveau intestinal, il remodèle l'identité transcriptomique des cellules caliciformes vers un état différencié fonctionnel, identifiant ces cellules comme cibles potentielles d'un mécanisme d'action intestinal indirect du tirzépatide dans le DT2.

Loss of one of the isoforms of the mesenchymal transcription factor PRRX1 in pulmonary fibrosis associated with systemic sclerosis: a new pathophysiological mechanism?

Agathe OK (1), Irene BASTIAN (1), Madeleine JAILLET (1), Audrey LACROIX (1), Nicolas POTE (1)(2), Bruno CRESTANI (1)(3), Arnaud MAILLEUX (1)

(1) Team iFi "Inflammation and Fibrosis in Respiratory diseases", UMR 1149 - Centre de Recherche sur l'Inflammation, Université Paris Cité, UFR de Médecine Site Bichat. (2) Service d'anatomo-pathologie, Assistance Publique-Hôpitaux de Paris, Hôpital Bichat, Paris, France. (3) Service de Pneumologie et Transplantation, Centre de Référence Constitutif des Maladies Pulmonaires Rares, Assistance Publique-Hôpitaux de Paris, Hôpital Bichat, Paris, France.

Systemic scleroderma (SSc) is a complex autoimmune disease resulting from an individual's genetic susceptibility combined with exposure to environmental factors. This pathology is first characterized by impairment of the blood vessels and immune system dysfunction, followed by progressive fibrosis which can affect the skin and internal organs such as the lungs.

SSc predominantly affects women (8 out of 10 patients are women), with a mean age of around 42 at the time of diagnosis. Patients with systemic sclerosis associated with diffuse infiltrative lung disease (SSc-ILD) have a reduced median survival. Currently, no anti-fibrotic or immunosuppressive maintenance therapy has been shown to improve survival. The only curative treatment remains lung transplantation.

The laboratory has identified the involvement of the PRRX1 transcription factor in the development of idiopathic pulmonary fibrosis (IPF). The PRRX1 gene encodes two forms of this protein: PRRX1a and PRRX1b. Both isoforms of the protein, PRRX1a and PRRX1b, are overexpressed in patients with IPF.

Surprisingly, our preliminary results show a loss of the PRRX1a isoform in the lungs of patients with ScS-PID. The aim of the project is to determine the role and consequences of the loss of the PRRX1a isoform in systemic sclerosis.

The PRRX1 expression profile was determined by immunohistochemistry (IHC – using an antibody directed against both isoforms) and by bioinformatic analyses of public scRNA-seq datasets from ScS-ILD. We used a siRNA that specifically depletes the PRRX1a isoform to investigate its role in the phenotype of primary lung fibroblasts derived from control patients (proliferation and myofibroblastic differentiation), comparing it with a siRNA targeting both isoforms.

PRRX1 is detected in vascular endothelial cells and lung fibroblasts from ScS-ILD patients by IHC, suggesting that these cells continue to express PRRX1b. Bioinformatic analyses revealed that PRRX1 expression in pericytes and smooth muscle cells is specific to ScS-ILD. Similarly to the concurrent depletion of both PRRX1 isoforms, the specific depletion of PRRX1a leads to a reduction in cell proliferation in lung fibroblasts from control patients. The differentiation of these cells into myofibroblasts is observed only in the presence of TGF- β 1.

In conclusion, PRRX1a exhibits 'profibrotic' functions in vitro similar to those observed following the simultaneous depletion of both PRRX1a and PRRX1b isoforms. The expression profile of the PRRX1 isoforms and their respective functions will be characterised in further detail using molecular tools, murine models of systemic sclerosis and precision-cutting lung slices (PCLS) from patients. A better understanding of these pathophysiological mechanisms will thus enable us to develop new therapies.

Role of Lcn2 in mast cells activation: Pathological implications in chronic kidney disease

Martin ZOUZOU, Université Paris cite, INSERM U1149, Centre de recherche sur l'inflammation, Laboratory of Immunology Nephrology and Cell Signaling (LINCS).

Chronic kidney disease (CKD) is a major and growing public health burden, with limited therapeutic options to prevent its progression. Renal inflammation and fibrosis are key drivers of disease progression, but the contribution of mast cells to these processes remains incompletely understood. Mast cells are innate immune cells that release preformed and newly synthesized inflammatory mediators following activation, particularly through the high-affinity IgE receptor FcεRI. Lipocalin-2 (LCN2), a protein strongly induced during kidney injury, has previously been linked to mitochondrial dysfunction and renal damage. Because LCN2 is highly expressed during kidney injury and is also expressed by renal mast cells, we hypothesize that LCN2 regulates mast cell activation and thereby contributes to CKD progression.

We are investigating IgE-dependent activation in wild-type and LCN2-deficient bone marrow-derived and peritoneal-derived mast cells. Mast cell responses are characterized by measuring calcium flux, degranulation and cytokine production. To identify the mechanisms underlying LCN2-dependent mast cell activation, downstream signaling pathways involved in FcεRI- and MRGPRX2-mediated activation, including STIM1, PLCγ, PKC, PI3K, and Akt, will be investigated together with mitochondrial morphology and function. In vivo, mast cell-specific LCN2 knockout mice will be studied in models of anaphylaxis and CKD. Finally, human kidney biopsies from patients with CKD will be analyzed to assess the association between LCN2 expression in renal mast cells, mast cell activation and renal lesions.

Preliminary data demonstrate that mast cell infiltration increases within the renal parenchyma of CKD mouse models. In addition, LCN2-deficient mast cells display reduced FcεRI- and MRGPRX2-dependent degranulation and cytokine release compared with wild-type cells. Therefore, we expect LCN2 to promote mast cell activation by regulating mitochondrial function and downstream signaling pathways. We also expect that mast cell-specific deletion of LCN2 will reduce mast cell activation, renal inflammation and fibrosis in vivo.

This project will define an unexplored role of LCN2 in mast cell biology and clarify how mast cell activation contributes to CKD progression. By linking LCN2, intracellular signaling, mitochondrial function and renal inflammation, this work may identify new therapeutic targets to limit mast cell-driven kidney damage and slow the progression of CKD.

Development of a preclinical model of carcinoid heart disease (CaHD)

Muyad AL HAFEZ (1), Jacques CALLEBERT, Antonino NICOLETTI, Giuseppina CALIGIURI, Jamila LASCHET

(1) LVTS, INSERM 1148, Team 1.

Carcinoid heart disease (CaHD) is a severe profibrotic complication of midgut neuroendocrine tumors, fundamentally driven by the hepatic metastatic bypass of serotonin (5-HT) first-pass clearance. The resulting systemic hyperserotoninemia targets the right heart chambers, where 5-HT binds to 5-HT_{2B} receptors on valvular interstitial cells, initiating rapid fibroblast proliferation and matrix deposition. The clinical advancement of targeted therapies is currently obstructed by a lack of appropriate preclinical tools; existing models utilizing systemic exogenous 5-HT, immunodeficient xenografts, or highly lethal continuous expression systems preclude longitudinal studies on the structural reversibility of the disease.

To overcome this methodological bottleneck, we developed a novel tumor-free, immunocompetent, and temporally controllable murine model of CaHD. We engineered a doxycycline-inducible Tet-On 3G lentiviral vector encoding the rate-limiting serotonergic enzyme, tryptophan hydroxylase 1 (Tph1), alongside a fluorescent mKate reporter. In vitro validation in the AML12 murine hepatocyte line confirmed stable genomic integration and robust, strictly doxycycline-dependent expression with zero basal transcriptional leakiness. RT-qPCR confirmed targeted Tph1 upregulation, which functionally translated into a significant 5-HT secretion (~35 nM) quantified via HPLC, without altering native host Ddc or Maa expression.

In vivo translation was achieved via hydrodynamic tail vein injection to selectively restrict vector integration to the abdominal venous drainage. While physiological mononuclear phagocyte clearance resulted in proviral DNA detection in the spleen, functional expression (native mKate fluorescence) remained strictly and exclusively confined to the hepatic parenchyma. The immunocompetent mice exhibited excellent systemic tolerance over a four-week longitudinal follow-up, presenting stable growth kinetics and fully preserved architectural integrity of the liver on H&E histological evaluation. Molecularly, continuous doxycycline induction drove a massive and highly significant upregulation of hepatic Tph1 ($p < 0.0001$). This was accompanied by a secondary, localized upregulation of hepatic Ddc ($p < 0.05$) and intestinal Maa ($p < 0.05$), indicating a robust physiological compensatory feedback loop in response to the engineered metabolic shift.

In conclusion, this first year successfully validates the genetic, functional, and spatial integrity of a highly specific, inducible hepatic serotonin-secreting platform. Once ongoing biochemical validations (plasma 5-HT, urinary 5-HIAA) are finalized, this model will be utilized to temporally map the onset of right ventricular dysfunction via longitudinal TAPSE echocardiography.

Characterizing viral load dynamics in community SARS-CoV-2, Influenza, and RSV infections : the role of age, sex, and vaccination

Laura MULAS (1), Maxime BEAULIEU, Antonin BAL, Florence DEBARRE, François BLANQUART, Jérémie GUEDJ

(1) IAME, INSERM.

Introduction – Acute respiratory infections (ARIs) caused by SARS-CoV-2, influenza viruses, and respiratory syncytial virus (RSV) remain a major public health burden. Since the COVID-19 pandemic, routine RT-PCR testing has become an integral component of respiratory viruses surveillance, generating large-scale datasets that include cycle threshold (Ct) values, an indirect measure of viral load.

In France, the RELAB network systematically centralizes these measurements from community laboratories, providing an unprecedented nationwide resource for studying within-host viral dynamics. Viral load kinetics are closely associated with transmission potential and disease severity, yet current knowledge relies mainly on small, intensively sampled clinical cohorts, limiting their generalizability to the wider population. Although mechanistic analyses of SARS-CoV-2 viral kinetics using routine RT-PCR data were developed during the COVID-19 pandemic, their application to seasonal respiratory viruses remain largely unexplored.

Objectives – We investigated whether modelling approaches developed during the COVID-19 pandemic could be adapted to routine post-pandemic surveillance of seasonal respiratory viruses (SARS-CoV-2, influenza A, influenza B, and RSV) to characterize viral kinetics and identify their main determinants from community-based RT-PCR data, despite these data typically being limited to a single sample per infection.

Methods – We analysed RT-PCR data from symptomatic SARS-CoV-2 (N=10,352), influenza A (N=8,891), influenza B (N=2,519), and RSV (N=4,916) infections collected through the French RELAB surveillance network over two winter seasons (2024-2026). For each virus, we fitted a target-cell-limited mechanistic model⁹ using nonlinear mixed-effects modelling (SAEM, Monolix). Covariate effects (age, sex, vaccination status, epidemic period and season) were selected using a stepwise BICc-based procedure and translated into differences in peak viral load and time to viral clearance through model simulations.

Results – Community-based RT-PCR surveillance enabled the characterization of within-host viral kinetics across SARS-CoV-2, influenza A/B, and RSV, despite relying on sparse, unrepeated sampling of symptomatic infections. Viral kinetics differed substantially with virus-specific effects of age, sex, vaccination status, epidemic period and season. Vaccination showed the largest covariate effect observed, although only for influenza B, reducing the estimated peak viral load by 1.4 Ct and time to viral clearance by 8.9 days. Age emerged as the most consistent determinant of viral clearance, although both the magnitude and direction of the effect varied across viruses. Seasonal and epidemic-period effects suggested differences in viral kinetics between circulating epidemics, particularly for IAV.

Conclusion – This study demonstrates that modelling approaches developed during the COVID-19 pandemic can be successfully adapted to routine community-based RT-PCR surveillance of seasonal respiratory viruses. Although our framework is limited by the observational nature of surveillance data, including potential biases, uncertainty in symptom onset, and the scarcity of repeated measurements, it identifies biologically meaningful determinants of viral kinetics and quantifies vaccination effects from routinely collected diagnostic data.

As multiplex RT-PCR surveillance continues to expand, this scalable approach could strengthen near real-time monitoring of respiratory viruses by identifying vulnerable populations, evaluating vaccination impact, and detecting changes in viral kinetics associated with emerging variants or shifts in population immunity.

Lung adipose metaplasia contributes to a fibrogenic and inflammatory microenvironment in Interstitial Lung Diseases (ILD)

Célia SCRIBE (1), Kinan EL HUSSEINI (1), Panagiota TSIRI (1), Camille CHENEAU (1), Madeleine JAILLET (1), Nicolas POTE (2), Pierre MORDANT (3), Raphaël BORIE (4), Aurélie CAZES (2), Bruno CRESTANI (4), Arnaud A. MAILLEUX (1)

(1) Université Paris Cité, INSERM, UMR 1149, Centre de recherche de l'inflammation, F-75018, Paris, France. (2) Service d'Anatomopathologie, AP-HP, Hôpital Bichat, F-75018, Paris, France. (3) Service de Chirurgie Vasculaire et Thoracique, AP-HP, Hôpital Bichat, F-75018, Paris, France. (4) Service de Pneumologie, Allergologie et Transplantation Pulmonaire, AP-HP, Hôpital Bichat, F-75018, Paris, France.

A common feature of Interstitial Lung Diseases (ILD) is the progressive remodeling of the alveolar niche, accompanied by aberrant mesenchymal differentiation. Alterations of the mesenchymal compartment include not only abnormal fibroblast differentiation but also adipose metaplasia. We identified white adipose tissue (WAT) in the remodeled peripheral subpleural regions of ILD lungs, whereas adipose tissue is normally restricted to the pulmonary hilum. These regions of adipose metaplasia are also enriched in immune cells. Intrapulmonary adipose tissue is frequently observed in advanced Idiopathic Pulmonary Fibrosis (IPF), the most common and severe form of ILD, and may influence the surrounding lung parenchyma through paracrine signaling. Previous work by the team demonstrated that adipocyte-conditioned medium decrease TGFb-induced myofibroblast differentiation through CRTP3-adiponectin receptor signaling pathway. However, adipocyte-conditioned medium also enhances fibroblast proliferation and migration, suggested that adipocytes maintain fibroblasts in a proliferative, undifferentiated state while promoting a profibrotic environment. In this project, we aim to elucidate the contribution of adipose tissue and its interactions with the lung parenchyma in the fibrotic process focusing on epithelial and immune compartment. A second objective of this project is to understand the cellular origin of adipocytes and identify the progenitor populations that give rise to white adipose tissue in the lung parenchyma. Conditioned media (CM) were generated from ILD adipose tissue explants (ExplantCM), as well as from cultured mature adipocytes (AdipoCM). These conditioned media were applied to several in vitro and ex vivo models, including THP1-derived macrophages, precision cut lung slices (PCLS) and analyzed by qPCR, ELISA and immunohistochemistry (IHC). Consistent with results in fibroblasts, ExplantCM induced a profibrotic response in PCLS, characterized by increased collagen and fibronectin at the mRNA and protein level. Regarding the immune compartment, ExplantCM increases the expression of pro-inflammatory cytokines (IL6, IFN γ), while reducing SPP1 expression, suggesting the induction of an alternatively activated macrophage phenotype distinct from profibrotic SPP1⁺ macrophages. In parallel, we observed enhanced adipogenic potential in IPF patients compared to controls, as evidenced by IHC showing an increased presence of ZNF423-positive cells, which is typically expressed in adipocyte progenitors. This project innovates by providing an in-depth characterization of pulmonary adipose tissue deposits in IPF and investigates its functional interactions with fibroblasts, epithelial cells and immune cells. These findings may contribute to a better understanding of the cellular and molecular mechanisms underlying ILD progression and reveal potential therapeutic opportunities.

Genome wide association study of model-derived time to viral clearance in COVID-19: comparison of methods in a simulation study and application to the discovery trials

Agl   PERRIN, Universit   Paris Cit  , IAME, INSERM, Paris.

Introduction – COVID-19 severity varies widely, from asymptomatic infection to life-threatening illness. Demographic and clinical factors explain only part of this variability, and genetic factors have been identified through GWAS and sequencing studies, particularly in type I interferon (IFN) pathways. Since viral load dynamics is associated with severity and remains highly variable even within homogeneous populations such as the DisCoVeRy 1 trial (NCT 04315948; EudraCT 2020-000936-23), we hypothesized that new genetic determinants of severity could be identified through GWAS of viral load dynamics. This work compares different GWAS approaches via simulation, with application to DisCoVeRy 1.

Methods – In DisCoVeRy 1, patients from the standard of care (SoC, N=329) and SoC+remdesivir (N=336) arms were regularly sampled for viral load. Among them, N=116 and 133 were genotyped (Illumina GSA array). Genetic data underwent quality control and imputation (Helmholtz-Munich server, 1000Genomes panel). The phenotype was individual time to viral clearance predicted from a target-cell-limited model, for SoC+remdesivir patients, the remdesivir treatment effect was set to 0 before calculation. We compared univariate and penalized regression, combined with different adjustments for population structure (principal component analysis (PCA) or ridge regression leaving one chromosome out) and variant selection strategies (Bonferroni correction or top 10 signals), yielding 14 method combinations. A hundred datasets were simulated under H0 (no genetic effect) and H1 (3 causal variants explaining ~30% of variability). For Bonferroni-based methods, we assessed FWER under H0 and power, precision, recall, and F1-score under H1. For top-10 methods, we assessed the proportion of datasets in which simulated SNPs were identified. Finally, we analysed DisCoVeRy 1 genetic and phenotype data, adjusting for age, sex, and type I IFN auto-antibody (AAB-IFN-I) status.

Results – After QC/imputation, 249 patients with 10,594,815 polymorphisms and AAB-IFN-I status were available. Bonferroni-based univariate regressions showed FWER >10%, versus ~0% for penalized regressions. After FWER correction, univariate regression with 3 PCA components achieved powers of 0.62, 0.19, and 0.05 (F1: 0.44, 0.13, 0.03) for the first, second, and third causal SNPs, penalized regression achieved powers of 0.33, 0.13, and 0.03 (F1: 0.32, 0.12, 0.03). Ridge regression leaving one chromosome out performed worse. Top-10 methods captured all 3 causal SNPs, with powers of 0.94/0.94 (SNP1), 0.74/0.80 (SNP2), and 0.39/0.52 (SNP3) for univariate/penalized regression, respectively. In DisCoVeRy 1, Bonferroni univariate regression identified no SNPs, while top-10 selection found 7 SNPs common to both approaches, with no overlap with previously reported loci.

Conclusions – While Bonferroni correction remains the preferred approach for GWAS, top-signal selection can provide valid results when sample size/power is limited. Univariate or penalized regression combined with PCA and top-10 selection proved practical for detecting associations in high-dimensional genetic data using a model-derived phenotype. Replication and meta-analysis are planned in DisCoVeRy 2 and confirmed variants will be integrated into the population viral load model.

Usage and regulatory acceptance of PBPK models in paediatric drug applications: a review of EMA and FDA public assessment reports

Chengli YU (1), Moreno URSINO (2), Sylvie RETOUT (3), Marylore CHENEL (4), Emmanuelle COMETS (1), Yumi CLEARY (5)

(1) Moclid, IAME, Université Paris Cité. (2) Université Paris Cité, HeKA, Inserm, Paris, France. (3) Institut Roche, Boulogne-Billancourt, France. (4) Pharmetheus, Uppsala, Sweden. (5) Pharmaceutical Sciences, Roche Pharma Research and Early Development, Roche Innovation Center Basel, Switzerland.

Introduction: The recent decade has seen considerable growth of paediatric physiologically based pharmacokinetic (PBPK) applications both in academia and industry, however, the number of regulatory submissions that includes paediatric PBPK modelling remains relatively small. The recent ICH M15 guideline provides a framework for Model Informed Drug Development (MIDD) evidence assessment which can be applied to examine the reliability of a PBPK model.

Objectives and Methods: To assess regulatory influence in decision making within the ICH M15 framework, we searched and reviewed European Public Assessment Reports (EPARs) and U.S Food and Drug Administration (FDA) public assessment reports that included paediatric PBPK modelling. 143 EPARs and 172 FDA public assessment reports published between 2015 and 2024 were systematically identified using the open-access database PaediatricData.eu and the commercial database PharmaPendium, respectively. A total of 25 distinct cases from EMA and 18 from FDA were kept after screening for the final review. Through those cases, we extracted information covering core elements of product characteristics, model attributes, context of use (CoU), modelling workflow and evaluation, regulatory conclusion, and ICH M15 assessment. Each application was subsequently classified according to the ICH M15 framework, including model influence, consequence of a wrong decision, model risk and model impact. The four MIDD assessment categories were scored as low/medium/high based on available information from reports. Each report was read by two readers and discrepancies between the assessments were resolved by consensus. Descriptive data analysis was performed in R. Comparative analyses between EMA and FDA regulatory practices were performed to identify trends in the acceptance and application of paediatric PBPK modelling.

Results: Applications involving extrapolation without paediatric observations were uncommon and showed lower acceptance rates, suggesting challenge of applying fully model-based approach to replace evidence from observed paediatric PK. Applications in children >2 years were more common than those in infants < 2 years, reflecting the greater physiological uncertainty and modelling challenges associated with younger populations. More cases with high model influence were accepted by FDA; approved applications were more associated with lower model impact, while rejected applications were more associated with higher model influence and model risk.

Conclusion: PBPK modelling shows promise to support paediatric clinical development, however, regulatory acceptance remains cautious.

Pitch Your Project

ABSTRACTS

Activable Antioxidant Microbubbles for the Treatment of Marginal Organs via Ex Vivo Perfusion

Berliana NOVIANITA (1)(2), Graciela PAVON-DJAVID (1)(3), Cédric CHAUVIERRE (1)(3)

(1) UMR-S U1148 INSERM, Laboratory for Vascular Translational Science (LVTS). (2) Galilée Doctoral School, Université Sorbonne Paris Nord. (3) Université Paris Cité, Université Sorbonne Paris Nord, F-75018, Paris, France.

In France, more than 23,000 patients are awaiting an organ transplant, and more than 900 die each year due to a lack of available organs. To expand the donor pool, Expanded Criteria Donors (ECDs), defined as individuals over 60 years of age or those aged 50–59 with significant comorbidities, are increasingly being considered for transplantation. However, organs from these donors, referred to as marginal organs, carry an increased risk of primary graft dysfunction and delayed graft function due to biological vulnerabilities, including oxidative stress, mitochondrial dysfunction, inflammasome activation, and endothelial dysfunction, which exacerbate ischemia-reperfusion injury. As a result, a significant proportion of these marginal organs are ultimately discarded.

Ex vivo organ perfusion opens a new therapeutic window to treat marginal organs and restore their biological function by enabling direct delivery of therapeutic agents to the graft at effective doses without systemic exposure of the recipient. This strategy overcomes the limitations of post-transplantation treatment, which is often constrained by limited graft penetration and the risk of systemic toxicity. It also enables continuous, dynamic assessment of organ function throughout treatment using perfusion parameters and perfusate biomarkers.

This PhD project aims to recondition marginal organs through ultrasound (US)-triggered, targeted delivery of antioxidants and nitric oxide (NO) gas using fucoidan-functionalized polymeric microbubbles during ex vivo perfusion, with the goal of restoring redox balance and mitochondrial homeostasis in the graft. Fucoidan functionalization enables preferential targeting of inflamed endothelial sites, while US triggers the localized release of therapeutic agents from the microbubbles and provides real-time, imaging-based monitoring throughout perfusion. The project will provide proof of concept for this multimodal therapeutic platform, which could ultimately contribute to reducing mortality among patients on the transplant waiting list by increasing the proportion of eligible organs obtained from ECDs and lowering the long-term healthcare costs associated with chronic diseases.

Study of the immunomodulatory role of IL-1RA in Pulmonary Fibrosis

Iris BOBINNEC (1), Julien TOURNEUR-MARSILLE (2), Ulrich MEINZER (2), Ibrahima BA (3), Caroline KANNENGIESSER (3), Raphaël BORIE (3), Bruno CRESTANI (3), Georges DOUMET HELOU (1)

(1) Junior group Helou, CRI, INSERM, France. (2) Viennois / Meinzer, CRI, INSERM, France. (3) Mailleux / Crestani, CRI, INSERM, France.

Idiopathic pulmonary fibrosis (IPF) is a progressive interstitial lung disease driven by aberrant wound healing and excessive extracellular matrix deposition, ultimately leading to respiratory failure. While the triggers of IPF onset remain unclear, inflammatory responses are increasingly recognized as contributors to disease progression and exacerbation. Among them, interleukin-1 (IL-1) signaling plays a central role in linking lung fibrosis to immune activation. This axis includes the IL-1 receptor antagonist (IL-1RA), encoded by the IL1RN gene and acting as an endogenous inhibitor of IL-1 receptor signaling, whose disrupted balance may drive chronic inflammation and fibrotic remodeling in IPF.

Overall aim: To investigate the role of IL-1RA in disease initiation and development, and the underlying mechanisms involved.

Objective 1: To assess the effect of IL-1RA deficiency on the initiation of pulmonary fibrosis in mice

We are comparing lung fibrosis development between IL-1RA knockout (KO) and wild-type (WT) mice using RT-qPCR, flow cytometry, cytokine quantification, and extracellular matrix deposition analysis. Initial results show increased pro-fibrotic genes and mediators in the lungs of IL-1RA KO mice, suggesting that IL-1RA deficiency may trigger spontaneous fibrosis — a finding that, if confirmed, would highlight the role of IL-1RA in disease initiation. Single-cell RNA sequencing on whole lungs will further help characterize the molecular and cellular basis of this phenotype.

Objective 2: To study the link between IL-1RA secretion and IPF severity

Previous studies suggest that IL1RN polymorphisms may alter its expression and increase susceptibility to IPF. We first stratified patients from our cohort based on IL-1RA levels in bronchoalveolar lavage (BAL). Preliminary data show increased IL-1RA levels and a higher IL-1RA/IL-1 β ratio in patients compared to controls, suggesting either a pro-fibrotic effect of IL-1RA or a compensatory secretion in response to excessive IL-1 β production. Further analyses will aim to correlate IL-1RA secretion with disease severity.

Objective 3: To understand the role of IL-1RA in macrophage-fibroblast crosstalk in IPF

Preliminary results show that treatment of human monocyte-derived macrophages with a fibrotic cocktail induces IL-1RA production in an IL-1 β -independent manner. Therefore, we are investigating the unknown factors that can trigger IL-1RA secretion in lung fibrosis. Additionally, we are assessing the effects of recombinant IL-1RA on primary lung fibroblasts (healthy donors vs. IPF patients), the impact of macrophage-secreted IL-1RA on fibroblasts in co-culture using a macrophage cell line deficient in IL1RN, and finally the role of recombinant IL-1RA in a 3D functional model (precision-cut lung slices) incorporating all pulmonary cell types.

By defining the role of IL-1RA in IPF pathogenesis, this project may uncover new mechanisms of disease and provide a foundation for the development of innovative therapeutic approaches.

Invasive pneumococcal disease dynamics among sickle cell disease children and potential benefit of next-generation pneumococcal conjugate vaccines: a 15-year national cohort study

Jee-Seon YANG, EPIC Team, IAME Laboratory, Université Sorbonne Paris Nord.

Introduction

Sickle cell disease (SCD) children are at increased risk of invasive pneumococcal disease (IPD), yet its recent post-COVID-19 epidemiology and the potential benefits of next-generation Pneumococcal Conjugate Vaccines (PCVs) remain unclear in this population.

Methods

We performed a quasi-experimental, population-based, interrupted time series analysis built from an exhaustive national surveillance system that includes SCD children living in France from 2010-2024. Primary outcome was IPD hospitalisation among SCD children. Trends in IPD incidence, risk ratios (RR) compared to general paediatric population, antibiotic resistance, and potential coverage of next-generation PCVs were analysed.

Results

Out of 18,596 SCD children included, 282 were hospitalized for IPD (21.4/10,000, 95CI% [19.0-24.1]). Median age for IPD was 4 years (IQR: 2-7). Cumulative incidence during the post-COVID-19 era was 41.9% higher (95CI% [22.1%-61.7%]) than the counterfactual no-COVID-19 scenario. Compared to the general population, IPD RR among SCD children did not decrease over time and was estimated to be 31.1 (95CI% [29.7-32.6]) during the post-COVID-19 era. Serotype distribution substantially varied from the general population with 9% of isolates covered by PCV15, 35% by PCV20 and 91% by PCV21, while pneumococcal polysaccharide vaccine PPSV23 only covered 41% among SCD children. Resistance to first-line empirical antibiotics (amoxicillin, third generation cephalosporins) remained <5%.

Significance

Despite specific anti-pneumococcal prevention strategies, IPD incidence increased recently among SCD children, involving serotypes that differed from the general population. PCV21 has the broadest serotype coverage and offers the opportunity to drastically reduce IPD burden in this high-risk population.

Development of a small perfusable artery *in vitro*

Arantxa VAQUERO (1), Emilie VONNA (1), Xiaochen HUANG (1), Teresa SIMON-YARZA (1)

(1) Team 3, Laboratory for Vascular Translational Science (LVTS), INSERM, Université Paris Cité Université Sorbonne Paris Nord.

Introduction

Arteries are blood vessels that transport blood from the heart to the organs. Their walls consist of three layers: the intima, media, and adventitia ; which provide both deformability and structural strength. When fabricating hydrogels for vessel-on-a-chip systems, Pullulan-Dextran polymers are used for their elasticity and flexibility properties, while the incorporation of Alginate enhances strength. This double-network approach creates a channel with mechanical properties similar to those of small arteries.

As part of a tissue engineering project aimed at creating a perfusable model containing blood vessels, our goal is to produce a biocompatible structure that mimic the physical, mechanical, and biological characteristics of a small artery, ultimately enabling its use as a model for studying vessel pathologies.

Materials and Methods

A biocompatible hydrogel composed of a cross-linked Pullulan-Dextran and Alginate double-network was molded to incorporate a 400 μm diameter channel, representing a small artery. The hydrogel was placed in a perfusion chamber made of a polydimethylsiloxane (PDMS) support screwed onto 3D-printed plates. A peristaltic pump perfused the microchannel to mimic physiological flow. A co-culture of smooth muscle cells and endothelial cells was introduced into the channel via microfluidics to mimic the structure of a small artery.

Results

The results demonstrated that the perfusion chamber generated shear stress within physiological ranges and produced a cyclic diameter elongation of 5% at a frequency of 1 Hz, which corresponded to the physiological frequency of heart beating. Moreover, primary endothelial cells from arteries were seeded to form a monolayer in the interior of the channel, in order to mimic the intima layer of arteries.

Conclusion

The development of this channel makes it a suitable model of an arterial vessel. Future work will focus on further developing it to study vascular pathologies, such as atherosclerosis.

Understanding the mechanisms of resistance to targeted internal radiotherapy in neuroendocrine tumors, and improving its efficacy

Lucy BOURLET (1), Killian VERON (2), Amytice TESSIER (3), Adel HAMMOUTENE (2), Nathalie COLNOT (4), Miguel ALBUQUERQUE (4), Jérôme CROS (1)

(1) Team PATH, Centre de Recherche sur l'Inflammation, Inserm 1149, France. (2) Team Sinkus, Centre de Recherche sur l'Inflammation, Inserm 1149, France. (3) Department of Nuclear medicine, Hôpital Beaujon, APHP Nord, France. (4) Department of Pathology, Hôpital Beaujon, APHP Nord, France.

Neuroendocrine tumors (NETs) of the pancreas and small intestine are considered rare tumors. Today, one of the treatments for these NETs is PRRT (Peptide Receptor Radionuclide Therapy). This technique uses a somatostatin analogue coupled to a radionuclide, Lutetium-177, which binds to a membrane receptor expressed by tumor cells (somatostatin receptor). The efficacy of this treatment varies significantly from one patient to another, with approximately 30% of patients being resistant. The goal of this project is to determine a molecular signature of sensitivity to PRRT, to understand the effects and mechanisms of resistance to radiotherapy, and to test new therapeutic combinations by associating them with potentiators, as well as to assess whether other targets could be used for PRRT.

We have assembled a cohort of patients with a NET of the small intestine or pancreas who received PRRT treatment. For certain cases, we also use their post-treatment samples. Tumor RNA was extracted for molecular profiling (RNAseq). The transcriptomic profiles of responders and non-responders will be compared to define a molecular signature of resistance. This will also help identify certain genes and signaling pathways that are key in resistance to PRRT.

In parallel, we are working with two NET cell lines, GOT-1 (small intestine) and NT18 (pancreas), which are exposed to increasing doses of PRRT to determine their sensitivity, and transcriptomes will be performed to determine the activated signaling pathways. We also expose these cell lines to DNA repair inhibitors (Nedisertib and Thiostrepton), which will then be combined with PRRT to evaluate their synergistic effect. Since cell lines do not account for the tumor microenvironment, we are carrying out these experiments on a fresh NET tissue slice model, which has the advantage of maintaining the tumor's architecture and cellular composition. After validation of the potentiators, we will test them in vivo on a murine model developed by Pr. Cherel's team in Nantes.

Less bugs, more signal: reporter species as microbiome biomarkers

Jorge MARTINEZ-GARCIA, EVRest (IAME), École Normale Supérieure.

Clinicians already rely on a handful of validated biomarkers to diagnose disease and monitor treatment response, without needing to measure every molecule in the body. Microbial communities carry out equally essential collective functions (biomass production, detoxification, resilience to antibiotic perturbation) but tracking these functions currently requires exhaustive, costly profiling of the entire community. This project asks whether a small, targeted subset of species, "reporter species", can play the same role for the microbiome that biomarkers play for disease: a minimal signature whose abundance is enough to diagnose community state and predict function, without full metagenomic sequencing.

To find these reporter species, we combine two lines of experimental work. First, we characterize a large library of bacterial strains one by one (how fast they grow, which nutrients they use, how resistant they are to different antibiotics, and what their genomes encode) building a trait profile for each species. Second, we assemble synthetic bacterial communities in the lab, starting from different combinations of species and environments, and expose them to antibiotic pulses while tracking, over time, which species dominate and how community-level functions change, such as resilience to the antibiotic, or the community's collective ability to break it down.

We then combine this experimental data with clinical cohort data, using both machine-learning and mathematical models of community dynamics to identify the minimal sets of species or genes that reliably predict these functions. The goal is a validated panel that can be tracked with fast, targeted assays such as qPCR, rather than full shotgun metagenomic sequencing, making it feasible to monitor microbial community health routinely, both in the lab and in the clinic.

A micropatterned hydrogel platform for human myocardium tissue modeling

Emma SULMONI (1)(2), Teresa SIMON-YARZA (1), Claire WILHELM (2)

(1) Laboratory for Vascular Translational Science (LVTS UMRS1148), INSERM. (2) Physics of Cells and Cancer (UMR 168), Institut Curie.

Cardiovascular diseases are a leading cause of mortality worldwide. In recent years, novel cardiac models have been developed for high-throughput assessment of therapies for heart diseases. Among these models, organoids and organ-on-chip technologies proved to be powerful in vitro models to study complex physiological processes in controlled environments. Current cardiac models face critical limitations, including insufficient vascularization, which compromises nutrient diffusion and waste removal. They also lack external stimulation mechanisms to enhance and synchronize cardiac contractions. Maintaining a physiologically relevant level of cardiac cells differentiation throughout the entire period of the experiment remains also a challenge.

We hypothesize that a microstructured 3D polysaccharide hydrogel will enable the engineering of highly differentiated, vascularized, and stimutable cardiac tissue.

We produced polysaccharide hydrogel composed of Pullulan, Dextran and Alginate. We designed various micropatterns on the top of the hydrogel matrix to investigate the impact of cell confinement on cardiomyocytes' ability to self-organize into microtissues, their survival and level of differentiation. The patterns were either circular or rectangular microwells with various dimensions. Cell experiments were conducted on collagen coated hydrogels to promote cell adhesion; controls were performed without collagen. Cell seeding on top of the hydrogel was first conducted with primary rat cardiomyocytes, at a density of 106 cells/hydrogel, to validate the collagen coating capacity to induce cell adhesion. Then, human induced pluripotent stem cells differentiated into cardiomyocytes (hiPSCs-CM) using small molecule activators of canonical Wnt signaling (GSK3 inhibitor) followed by a small-molecule inhibitors of Wnt signaling (Porcupine inhibitor) were seeded at a density of 3.105 cells/hydrogel. Cardiomyocytes survival was analyzed by beating observation throughout 7 days. Cellular organization was assessed by immunofluorescence.

Hydrogels matrices with various preserved pattern geometries on the top were produced. Rat cardiomyocytes adhered to the patterns of the hydrogels coated with collagen independently of the pattern design. The cells arranged in monolayers and tended to form branching structures between the walls of the microwells. In the uncoated hydrogels, the cells formed spheroids for all patterns.

hiPSCs-CM seeded on collagen-coated hydrogels tended to form small spherical aggregates, independently of the pattern. Beating of the cells was observed from 2 days after seeding until the end of the culture experiment at day 7 suggesting a good preservation of the differentiated state of the cardiomyocytes. It is important to note that, due to limited availability, the hiPSCs-CM were seeded at a lower density than the rat cardiomyocytes and this may have affected their reorganization in the patterns. The experiment should be performed again at higher cell density to assess the impact of cell concentration on the 3D cell arrangement of hiPSCs-CM.

Next experiments will focus on improving the level of differentiation of the cardiac model by incorporating cardiac fibroblasts cells and endothelial cells for enabling vascularization. Then, the model will be magnetically stimulated and perfused.

This model can be used for drug screening of heart therapies while integrating potential implications for personalized cardiovascular treatments.

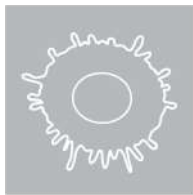
Sponsors

Reliable data starts with the right cell culture tools



Whether you're preparing your first poster, your next publication, or the next step in your project, you need data you can trust. With human primary cells, optimized media, and expert support, PromoCell helps you generate physiologically relevant results with confidence.

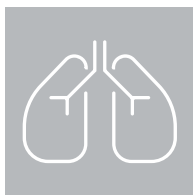
Advancing research with cells, optimized media, and expert support



Immunology research



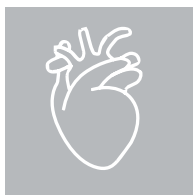
Kidney research



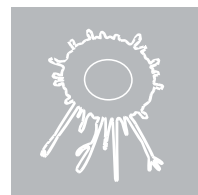
Respiratory research



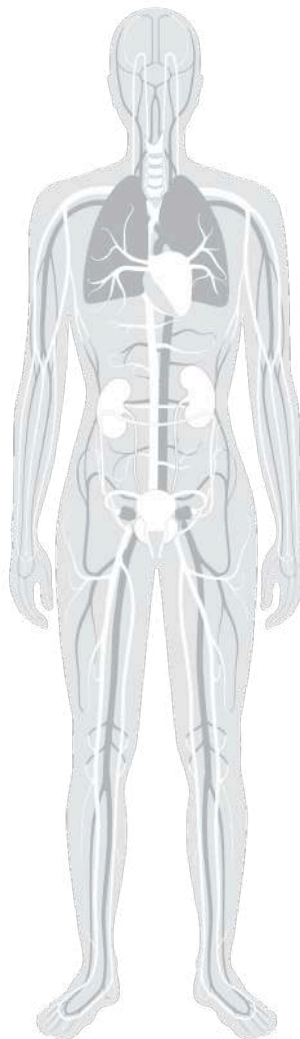
Dermatology and skin
biology



Cardiovascular research



Immuno - oncology



Why visit PromoCell?

- Generate more physiologically relevant data with human primary cell models
- Improve data reproducibility with optimized media tailored to each application
- Simplify donor selection with fully documented HLA-typed cells
- Customize media formulations to your research needs
- Save valuable lab time with our cell expansion service
- Talk through your project with our cell culture specialists



Let's find the right cell culture solution for your project



Meet our experts, discuss your research challenges, and find the cell models and media that fit your project.

Scan the QR code to schedule a meeting with us or visit us during the 6th edition of the Bichat Beaujon Young Researchers' Day (JCBB).

Your science. Our products. Your progress.

www.promocell.com | info@promocell.com | ©PromoCell GmbH



When you need it all.

300,000+ citations of Proteintech products demonstrates what matters most: the success of its reagents in the hands of scientists.

Proteintech Product Range

- Primary antibodies against 13,500+ human targets
- **humanokine** GMP recombinant cytokines and growth factors
- **chromotek** nanobody reagents
- **FlexAble** antibody labeling kits
- ELISA kits
- Magnetic cell separation systems
- Tag / Control antibodies
- CoraLite fluorescent-dye conjugated antibodies
- SignalBright chemiluminescent substrates
- NeutraKine neutralizing antibodies
- Tagged immunogens
- Secondary antibodies
- Phantom viability dyes
- Prestained protein markers
- IHC kits

Visit **ptglab.com** to browse the full product range.

When you need it all.

ptglab.com



Spectral Flow Cytometry Meets Modularity

Pair Your CytoFLEX LX/S* Flow Cytometer with the CytoFLEX mosaic Spectral Detection Module to Enable Your Spectral Flow Cytometry Capabilities

The CytoFLEX mosaic Detection Module is the natural evolution of our CytoFLEX Platform. A smart approach to spectral, it connects to your CytoFLEX LX/ S Flow Cytometer, giving you the flexibility to mix and match different detection methods in one instrument.

The Best CytoFLEX You Can Get



**CytoFLEX mosaic 88
Detection Module for
CytoFLEX LX Flow
Cytometer**
(88 channels of detection:
1 FSC, 6 SSC, 81 fluorescent)



**CytoFLEX mosaic 63
Detection Module for
4-laser CytoFLEX S Flow
Cytometer**
(63 channels of detection:
1 FSC, 4 SSC, 58 fluorescent)



A flexible approach to
spectral flow cytometry



Data you can trust



Makes the complex easy

Bring radiance to your panels

BioLegend from Revvity is the leading provider of flow cytometry reagents, bolstered by an extensive antibody clone library, spectrum-spanning fluorophores, and an unwavering commitment to quality. Our dedicated scientists support you at every step of your workflow, helping you to construct the perfect panel.

A full workflow of reagents

- New antibody conjugates to expand your flow panels, including Spark PLUS V450, B488, B574, and YG581, and StarBright UV575, 605, 665, 740, and 795.
- CytoScribe™, our cloud-based software, provides researchers with free accounts to analyze their flow cytometry data.
- Pre-optimized 14+ color panels provide consistent results for Treg, T cell, and myeloid cell phenotyping.
- Our Compensation Beads bind to mouse, human, rabbit, rat, hamster, and donkey Ig isotypes.

Discover the difference at biolegend.com

revvity





Fisher Scientific : votre partenaire au service de la science

Fisher Scientific, marque de Thermo Fisher Scientific, accompagne au quotidien les laboratoires et les acteurs de la recherche en leur donnant accès à une offre complète de produits, d'équipements et de services scientifiques.

Grâce à un large portefeuille associant les solutions des marques de Thermo Fisher Scientific à celles de nombreux fournisseurs de référence, Fisher Scientific répond aux besoins des laboratoires dans de multiples domaines : consommables, équipements, produits chimiques, sciences de la vie, sécurité et services.

Au-delà de l'étendue de son offre, Fisher Scientific simplifie l'approvisionnement des laboratoires grâce à son expertise scientifique, ses solutions digitales et ses services de gestion des achats et des approvisionnements. Notre ambition : permettre aux scientifiques de se concentrer sur l'essentiel — faire avancer la science.

Votre contact: Alain Nardella 06 72 28 02 69 ou alain.nardella@thermofisher.com



Cellimaging

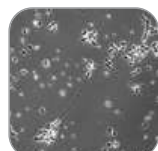
Simple yet powerful

Imaging made easier with EVOS imaging systems

Invitrogen™ EVOS™ imaging systems are designed to help eliminate the complexities of high-end microscopy without compromising performance. Exceptionally versatile, intuitive EVOS microscopes are excellent for a broad range of imaging applications—including cell culture, time-lapse imaging, and high-resolution image capture from slides, dishes, flasks, and microplates. EVOS systems deliver publication-quality images and data in no time, every time, at an exceptional value.

Key features:

- Publication-quality imaging—designed with top-of-the-line imaging components
- Compact, efficient design—developed with scientists' workspace and workflow needs in mind
- Intuitive software—equipped with onboard software to view, capture, and process images quickly and easily



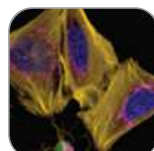
Cell culture



Tissue sections



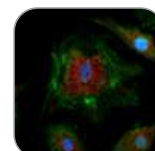
Cell viability



Cell death



Cell structure



Cell proliferation





MEET NOVOGENE
EUROPE AT JCBB

YOUR TRUSTED MULTI-OMICS PARTNER



GENOMICS



TRANSCRIPTOMICS



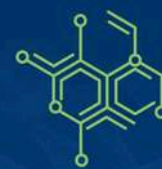
EPIGENOMICS



SINGLE-CELL & SPATIAL



PROTEOMICS



METABOLOMICS

Local labs. Local support. Global scale.

[EU.NOVOGENE.COM](https://eu.novogene.com)



Search 'Novogene Europe'



Scan to explore
our capabilities

Innovation
Fellowships
Conferences

Trainings
Equipment grants
Structuring projects

The BioConvergence pour la Santé (BioConvS) initiative aims to position the Île-de-France region as a key player in **synthetic biology, bioproduction, and biotherapy**. By catalyzing interactions, funding cutting-edge equipment and funding interdisciplinary projects, BioConvS seeks to make advanced technologies for life design and engineering accessible to a broad community of researchers in Île-de-France. We support the development of projects along five main research axes:

1. Bioengineering and bioproduction
2. Development of therapeutic proofs of principle
3. New, high throughput and standardizable analytical methods
4. Digital or AI tools to improve data analysis or modeling in quality control and production methods
5. Ensure a responsible, ethical and inclusive Bioconvergence

Our actions

Yearly calls for PhD fellowships and equipment

Every year we support **PhD students** who work in Ile-de-France on one of the DIM's thematics. We also open yearly calls for small, medium or heavy equipment grants.

Trainings

We sponsor and organize **trainings** with a specific focus on bioproduction, biotherapies, innovation and entrepreneurship. Follow our newsletter to be kept informed about our offers!

Conferences

We sponsor and organize **conferences** and events in Île-de-France on the thematics of the DIM. We also organize a yearly **symposium** for the members of our community, as well as smaller **scientific events**.

Apply for a
conference
funding:



A Digital Platform for Collective Intelligence

We want to promote fruitful interactions between members of the BioConvS community in order to:

- accelerate the search for **partners** necessary for setting up projects
- catalyze the **exchange** of ideas and best practices
- facilitate access to **funding** through calls for projects.

The platform will allow organizations to make themselves known within the community, by sharing their strengths, projects, areas of expertise and news.

<https://bioconvs.org>



subscribe



*Le DIM BioConvS est une action financée
par la Région Ile-de-France*

contact@bioconvs.org

follow



Acknowledgments

The JCBB organizing committee warmly thanks all participants, young researchers and speakers for joining us and contributing to this year's meeting through their research, expertise and enthusiasm.

We are grateful to the research units, institutions and partners supporting this event and fostering scientific exchange among young researchers.



A special thank you to our main sponsor

We warmly thank PromoCell for its generous support in making this edition of the JCBB possible.

We are also grateful to all our sponsors and partners for their valuable support:



Domaine de Recherche et d'Innovation Majeur

Finally, we would like to thank the entire organizing team for their time, energy and dedication in bringing this event to life.

We look forward to sharing a day of science, discussion and connection with you!

